

ASSETOPS

Integrated Product Experience and Canonical Screens

v6.9 – Freeze: Commit releases sealed source envelopes through normal ingestion (never writes Site history); Draft · Paused canonical example; staging vs submitted gateway output; Expectations as private test-oracle metadata

Core proposition. AssetOps turns fragmented telemetry, operational records, configuration history and field evidence into a ranked view of what is healthy, where money or service is at risk, what should happen next, how confident the conclusion is, and whether the intervention actually worked.

10 September 2026 · supersedes v6.8 and earlier · FREEZE

1. What this revision fixes

v6 keeps the v5 architecture (stable global menu, contextual entity tabs, one archetype-specific subtab row, simulator discipline, incident/work/verification model, evidence-readiness model) and corrects the places where v5 was inconsistent with its own rules. All 33 mockups are regenerated from a single screen generator so that one shell, one status vocabulary and one KPI grammar apply everywhere.

Problem in v5	Resolution in v6
Site tabs <i>Findings</i> , <i>Work</i> , <i>Financials</i> and <i>Evidence</i> share names with left-navigation items, and the document only explained the rule for Evidence and Financials. Site Findings/Work were undefined and risked becoming second lists with their own states.	Explicit scoped-index rule (§2.3): a site tab that names a global object class is a read-only, site-filtered index with the same columns, states and lenses, no KPI strip, and a <i>Manage in <Global></i> link. Two new screens (8.20, 8.21) show the rule. Site-owned content (readiness, performance, claim ceilings, technical bridge) lives only under the site.
Left navigation listed <i>Opportunities</i> and <i>Risks</i> as separate items although the document defined them as lenses over Findings, violating “left nav highlights the object class”.	Opportunities and Risks become the subtab row of <i>Findings</i> (All Opportunities Risks Awaiting verification). Left navigation shrinks to 10 object classes / roll-ups.
Site <i>Work</i> tab bundled Actions, Incidents and Maintenance with no defined structure.	Site Work uses one subtab row that mirrors the WORK group of the left navigation one-to-one: Actions Incidents Maintenance, with counts in the labels.
Asset tabs used <i>Maintenance</i> while site tabs used <i>Work</i> for the same scoped concept.	Asset tab renamed <i>Work</i> ; scoped-index tabs always carry the global object-class name.
Maintenance work requests had both <i>Impact</i> and <i>Verification</i> tabs; Actions had only <i>Verification</i> .	All managed tasks (Actions and Maintenance work requests) share six tabs: Overview Plan Execution Evidence Verification History. Impact content sits inside Verification as the “measured impact” panel.
Finding <i>Overview</i> repeated the evidence chain, alternatives and recommendation that have their own tabs.	Overview states the bounded claim, consequence, confidence and claim boundary, and routes to the Evidence, Alternatives and Recommendation tabs.
Simulator controls highlighted <i>Portfolio</i> in the left navigation although scenario controls must live outside operator pages.	Simulation controls open from the workspace switcher with no left-nav highlight; provenance (SIMULATED + run ID) is carried by the shell, not by page content.
SIMULATED badge, breadcrumbs and time window were placed inconsistently; several screens had truncated or overlapping labels (“NEEDS REVIE”, “VERIFICATION WIN”, overlapping action names, timeline labels).	One shell: portfolio switcher, search, time window, SIMULATED badge and user in the top bar; breadcrumb above every title; tab row; optional subtab row; chip row; KPI strip; panels. All text is ellipsis-safe and timeline labels alternate rows.
KPI cards were coloured arbitrarily (e.g. a neutral fuel-opening figure in blue) so colour did not carry meaning.	Only assessment cards are coloured (Healthy / Watch / Needs attention / Degraded / Unknown). Metric cards are neutral and always state their basis (measured / modelled / derived).

Status words varied by screen (Investigating, Work open, Verification pending, Completed, Due...) without a defined vocabulary.	Single status vocabulary (§5) for assessments, confidence, finding states, managed-task states, incident states, evidence status and basis.
Global menu had <i>Reports</i> with no definition.	Reports defined as scheduled/exported summaries carrying provenance and basis labels; no analytical status of their own (P1).

1.1 v6.1 changes (after v6 review)

Issue found in v6	Resolution in v6.1
Portfolio panel titled “Ranked portfolio actions” while its rows are Findings.	Renamed “Highest-priority findings”; an Action column shows the accepted Action (or “Accept recommendation”) per row.
Findings subtab row mixed interpretation lenses (Opportunities, Risks) with a workflow state (Awaiting verification).	Lenses row = All Opportunities Risks only; workflow state is a filter chip. Same on the site-scoped Findings tab.
Site Health listed the Gateway next to physical assets as if “Online” were equivalent to “Healthy”.	Site Health split into two blocks: Physical asset condition (assessment vocabulary) and Instrumentation / source health (Online / Stale / Offline + quality grade).
Battery Health cards used local words (Normal, Stable, Uncertain) that could be read as an alternate status enum.	§5 now states that the canonical assessment vocabulary applies to overall assessments only; sub-indicator cards may use domain-specific condition descriptors and are labelled as such on the screen.
Site Financials carried an overall “Needs attention” assessment with no deterministic financial rule.	Financial pages carry no assessment. Header shows “Financial consequence identified: n lines” plus confidence; status stays on the source findings.
Action Verification showed State: Verification window together with Outcome: Verified resolved.	State: Verification window · day 5 of 7; Outcome: Provisional pass. Final Verified / Ineffective / Inconclusive is set only when the window closes.
Action, Incident and Maintenance detail used different header fields.	Common managed-object header on all three: Owner Due / SLA State Source Success / closure criterion.
Warranty tab and Reports item looked implemented.	Warranty marked P1 (deferred) in the asset tab set and rendered as “Warranty (P1)”; Reports marked P1 in the navigation table.

1.2 v6.2 changes (simulation architecture)

v6.1 position	v6.2 position
Simulation was a workspace property: a simulated portfolio, a Simulation workspace controls page, a SIMULATED badge in the top bar of every page.	Simulation is a source mode of a Site . A simulated Site is a normal Site object whose evidence is produced by a versioned SimulationRun. Live and simulated sites coexist in one portfolio; provenance is shown in the site header (and on objects derived from that site); the top bar only flags a wholly simulated workspace.

No model for how simulations are authored, versioned or rerun.	New simulation objects (§3.1): ScenarioTemplate → ScenarioDefinition → SimulationRun → evidence for a Site (v6.2 said “→ simulated Site”; superseded by v6.4); SimulationTruth as a private oracle. Three modes: Synthetic site, Scenario clone, Historical replay.
Screen 8.12 showed simulator controls.	Screen 8.12 is Scenario Lab (template-first authoring, events vs evidence conditions, expectations as assertions). New Simulated site in Replay mode and Truth vs observed (admin only) screens (now 8.16, 8.17).
No simulation entry points defined.	Entry points, not navigation items: Sites › + Add site ▼ (Register live / Create simulated / Clone into scenario); workspace utility menu › Scenario Lab / Manage runs; simulated-site header › Replay, Scenario definition. Left navigation unchanged.

1.3 v6.3 change

(Superseded in part by v6.4 – see §1.4.) Added §3.2 *Simulation execution contract*: (1) Site ↔ SimulationRun identity and lifecycle (v6.3 wording “one run → one new immutable Site ID”, now replaced by stable declarative identity); (2) ingestion equivalence (same source envelopes, mapping IDs, quality semantics, no simulator-only path into derived analytics); (3) provenance propagation to every descendant including Financials, VerificationOutcome and exports; (4) Replay as an as-of view served from the versioned run snapshot. Two acceptance tests added. No screen changes.

1.4 v6.4 changes (frozen Simulation Integration Direction)

v6.4 is a targeted product-model revision. It does not reopen navigation, screen grammar, status vocabulary, claim discipline, scoped/global object identity, Recommendation / Action / Verification semantics, the R2/R5 boundary or the truth boundary.

Group	v6.3	v6.4
Identity / lifecycle	Each SimulationRun created a new simulated Site with its own Site ID; reruns created new Sites.	A Site has a stable canonical <code>site_id</code> declared in its Site definition (Site YAML for the MVP) and carried through all source and product data. SimulationRun references a Site and never allocates identity. Backend Site registration is deferred (not P0).
Site history / time	Site history was implicitly one run; Replay was run-centric.	Normal historical context is Site + selected time window . All tabs resolve against it. Artifact time semantics are defined (point-in-time, interval, time-valid configuration, lifecycle).
Simulator workspace	Scenario Lab authored template overlays and produced a Site per run.	Two setup paths: define a simulated Site from a template (canonical Foundation schema incl. <code>site_id</code>) or use an existing Site; then choose scenario + simulated interval and run. Developer run-management surface added (8.15). Interactive simulator mode defined with the frozen causality chain (made canonical, not future, in v6.7).
Ingestion	Same source envelopes; no simulator-only path into analytics.	Strengthened: serialization/persistence is part of the canonical path; the product must be reproducible from persisted source envelopes without simulator runtime objects or truth; file-backed is acceptable, a network hop is not required.

Protocol / persistence	—	Protocol-adapter direction (CAN, Modbus, MQTT, HTTP, serial, file) feeding the same canonical envelopes; persistence principle “production-compatible seams, MVP-simple infrastructure”; MVP stack note.
Provenance / Replay	Provenance inherited by descendants; Replay = as-of view of one run snapshot.	Provenance adds canonical site_id and source mode; multi-run windows stay traceable per contributing run; run_id is never Site identity. Replay = as-of view of recorded Site history; run is provenance/filter/admin context.
Acceptance	Tests required a new Site per run.	Replaced by tests A-H: declarative identity, multiple runs one history, run ≠ Site identity, normal ingestion, truth isolation, interactive causality, time-window consistency, overlap protection.

v6.4 supersedes these v6.3 statements: (i) every SimulationRun necessarily creates a new Site ID; (ii) run identity as the normal historical/product boundary; (iii) a run-addressed Site experience as target architecture; (iv) the Sites index treating runs as Sites; (v) “one run → one immutable simulated Site; Clone & rerun creates a new Site”; (vi) “deleting a run archives its Site”; (vii) Replay served from a single run snapshot as the primary address; (viii) the v6.3 test “create a second run and get a second Site ID”. Historical change tables in §1.2–1.3 are retained as record and marked where superseded.

Unresolved decisions (explicitly guarded, not decided here): branch/version-selection model for overlapping runs on the same Site/time range; backend Site registration flow and its acceptance rules; whether a development-only “replace evidence stream” mode is still needed once registration exists; protocol-level fidelity requirements per adapter.

1.5 v6.5 changes (Simulator Lab mockup integration)

v6.5 integrates the Simulator Lab mockup and description. It changes only the simulator workspace and its contracts; Site identity, Site + time-window history, ingestion equivalence, provenance, Replay, truth isolation and all operator-product semantics are unchanged.

Area	v6.4	v6.5
Workspace	“Scenario Lab” as a workspace utility with no shell of its own.	Simulator Lab: a separate developer workspace with its own shell and navigation (Home Sites Simulator Lab Scenarios Site Templates Library Documentation Settings). Scenario Lab becomes its Scenarios area. Operator navigation unchanged; the bridge is <i>Open in AssetOps</i> (simulator → product) and <i>Open in Simulator Lab</i> (product → simulator).
Purpose statement	Simulator emits canonical envelopes.	Simulator Lab is a virtual physical site that behaves like a real site and publishes through production ingestion semantics; it owns world, environment, controllers, devices, sensors, events, time and gateway publication (§3.9).
Simulation time	Simulated clock listed as a run attribute.	First-class time semantics: simulation window, timestep, execution mode (real time / accelerated / max), simulated clock/elapsed/progress vs wall-clock; per-schedule cadences anchored to the simulation clock; Pause / Step / Fast-forward / Reset; execution speed never changes results (control names finalised in v6.7).

Devices and gateway	Truth vs observed only in the AssetOps admin overlay.	Devices & Sensors panel (truth vs reported, device realism) and Gateway & Ingestion panel (exact published envelopes, realistic ingestion conditions, Inspect raw gateway output) in the simulator. The AssetOps admin overlay reads the same truth.
Screens	8.12 Scenario Lab; 8.13 Replay; 8.14 Truth; 8.15 Runs.	8.12 Simulator Lab Site View; 8.13 Devices & Sensors / Gateway & Ingestion; 8.14 Scenarios authoring; 8.15 Runs; 8.16 Replay (AssetOps); 8.17 Truth vs observed (AssetOps admin). Later screens renumber +2.
Acceptance	A-H.	Adds I (execution-speed invariance) and J (simulator surfaces contain no product conclusions).

Superseded v6.4 wording: “Scenario Lab (authoring) – workspace utility menu, no left-nav highlight” becomes “Simulator Lab workspace – separate shell”; “Scenario definition ↗” link on simulated sites becomes “Open in Simulator Lab ↗”. No other v6.4 rule is superseded.

1.6 v6.6 changes (feature gating and single line diagram)

(1) The Simulator Lab is a **gated feature**: enabled or disabled by a deployment-wide flag now, per tenant later (§3.10). (2) The Site View is a single line diagram with breakers, PCS, AC-bus voltage/frequency and unserved load (8.12). (3) Acceptance test K added. Nothing else changes.

1.7 v6.7 consistency pass (after simulator review)

#	Issue	Fix
1	§3.6 called the interactive simulator “future / not P0” while §3.9 and 8.12 defined it as the current Simulator Lab.	§3.6 rewritten as <i>Interactive simulator runtime and causality</i> ; the SLD + world interaction + time engine + devices/sensors + gateway output is the canonical simulator experience.
2	Screen numbering drift (Runs after Truth).	Order fixed: 8.12 Site View, 8.13 Devices & Gateway, 8.14 Scenarios, 8.15 Runs, 8.16 Replay, 8.17 Truth comparison.
3	8.14 Scenarios was rendered inside the operator shell.	8.14 and 8.15 now use the Simulator Lab developer shell.
4	Truth visibility statements contradicted each other.	Invariant restated in §3.9: truth may appear anywhere inside Simulator Lab; never in envelopes, evidence, operator UI or exports. Test J rewritten.
5	Restart conflicted with immutable runs.	Draft / Commit / Rerun / Replay semantics defined (§3.9); control renamed Reset (draft); §3.2 rule 1 says immutable once committed.
6	Injected events broke scenario+seed determinism.	Determinism identity includes the ordered intervention log (§3.6); acceptance test L.
7	Header arithmetic inconsistent.	Example now 2d 14h 32m = 37% of 168 h, wall 00:37:32 at 100×; label reads Current speed.
8	Jump-to implied skipping ticks.	Renamed Fast-forward to simulated time.

9-10	Gateway table under-represented the envelope; timestamp wording vague.	Summary table vs Inspect raw hierarchy stated; timestamp vocabulary frozen: source_time, gateway_time/published_at, received_at; Replay keyed on received_at.
11	“Scenario definition ↗” survived in the provenance strip.	Replaced by Open in Simulator Lab ↗ (deep-links to scenario/run).
12-13	“Scenario Lab” in the pipeline statement; stale nav-invariant wording.	Pipeline reads Simulator Lab / Scenarios → ...; invariant scoped to the operator shell.
naming	Simulator “Sites” could be read as a second Sites index.	Defined as site definitions available to the simulator, same site_id, no simulator-owned identity.

1.8 v6.8 freeze-candidate pass

Issue (v6.7 review)	Fix
SimulationRun still “immutable” in §3.1.	§3.1 row: time-bounded execution record; mutable only while Draft, immutable once Committed; two status dimensions.
SimulationTruth row kept the old “readable only by the admin overlay” wording.	§3.1 row aligned with the Simulator Lab truth rule: private to the simulator/evaluation domain; surfaced inside Simulator Lab, the Expectations layer and the admin overlay; never in envelopes, Evidence, analytics, operator UI or exports.
8.14 said Run creates an immutable run.	Run creates a Draft run executing into its staging stream; Commit makes it immutable.
Commit had no UI action.	<i>Commit run</i> added to the Simulator Lab top bar beside Open in AssetOps (8.12/8.13) and to the Runs table, visible only for eligible Drafts (§3.9).
Open in AssetOps offered on a Draft run.	Disabled until Commit with an explicit note; Runs table shows it only for Committed runs.
Run status vocabulary mixed lifecycle and execution.	Lifecycle: Draft Committed. Execution: Ready Running Paused Completed Failed Blocked. Separate columns and badges.
Test I used the old scenario+seed shorthand.	Test I now references the §3.6 deterministic run identity.
8.13 “only content AssetOps can ever see”.	Reworded: the only simulator-produced information that crosses into normal ingestion; AssetOps derives from envelopes, not the panel.
received_at could look simulator-generated.	§3.9: simulator shows source_time and gateway_time before hand-off; received_at is recorded by the ingestion boundary and visible only after receipt.
Boundary sentence implied the admin overlay was an exception.	“Normal AssetOps ingestion and the operator experience see only published observations...”; admin overlay stated as out-of-band.

1.9 v6.9 freeze pass (after v6.8 review)

Priority	Change
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P0	Commit rewritten: seals and persists the staged canonical source envelopes and releases them through normal ingestion; Site history is populated only by validation → normalization → Evidence; Commit never writes Site history or derived objects (§3.9, 8.14).
P0	Canonical example run is Draft · Paused · 37% on 8.12/8.13, so Commit run is legitimately eligible; control row shows Resume.
P0	8.13 gateway output is state-sensitive: Staging (Draft) / Submitted to AssetOps (Committed) / Ingested by AssetOps; Draft view shows “Destination: AssetOps canonical ingestion · release on Commit” and no received_at.
P0	Expectations rule frozen (§3.6): private test-oracle metadata, never pipeline input; Test J rewritten; 8.14 badges the panel PRIVATE TEST ASSERTION – NOT PIPELINE INPUT.
P1	Commit removed from the 8.14 Scenario header (Commit belongs to a run); header is Open target Site ↗ + Run v4 ►.
P1	List-level Commit and Open in AssetOps removed from the 8.15 Runs header; Commit remains a row action for eligible drafts and a header action on a specific Draft run.
P1	§3.1 entry points: “Open in AssetOps on every run” replaced by “on an eligible Committed run”.
P2	SLD labels truth vs reported explicitly (18.4 kW true · 18.2 kW reported; 274 L true · 289 L reported); legend: primary values = simulator truth, reported values labelled.
P2	received_at exists only after ingestion; Draft Inspect raw carries none.

With these changes the document is **frozen** as the product-design baseline. Subsequent work is task decomposition (Spec Partner), not specification change.

2. Canonical product architecture

Navigation group	Global menu items	Root object / workspace
Operate	Portfolio Sites Assets	Portfolio roll-up; Site; Asset
Intelligence	Findings (lenses: All Opportunities Risks Awaiting verification)	Finding
Work	Actions Incidents Maintenance	Action; Incident; Maintenance work request
Assurance	Evidence	Portfolio evidence roll-up
Business	Financials Reports (P1)	Portfolio financial roll-up; scheduled/exported summaries with provenance and basis labels, no analytical status of their own

Navigation invariant. The left navigation highlights the class of the current root entity or workspace; contextual tabs describe aspects of that entity; at most one additional subtab row is allowed. Deeper detail uses sections, drawers or linked objects, never another permanent tab level. This invariant governs the operator shell; the Simulator Lab is a separate shell with its own navigation (§3.9).

2.1 Canonical tab sets

Root object	Left nav	Contextual tabs	Allowed subtab row
Site	Sites	Overview Foundation Health Performance Findings Work Financials Evidence	Foundation: Definition Topology Controls Changes Readiness · Performance: archetype domains · Findings: lenses · Work: Actions Incidents Maintenance · Financials: Overview Drivers Assumptions Scenarios
Asset	Assets	Overview Health Performance Configuration Work Evidence Warranty (P1, deferred – no content contract yet)	Work: Actions Maintenance
Finding	Findings	Overview Evidence Alternatives Recommendation History	— (drawer for claim lineage)
Action	Actions	Overview Plan Execution Evidence Verification History	—
Maintenance work request	Maintenance	Overview Plan Execution Evidence Verification History	—
Incident	Incidents	Overview Timeline Findings Work Evidence History	Work: Actions Maintenance
Simulated site	Sites	Same as Site	Same as Site; Replay is a mode, Open in Simulator Lab is an outbound link
Portfolio	Portfolio	Command-center lenses	—
Portfolio evidence / financials	Evidence / Financials	Roll-up views	Filters only

2.2 Archetype-specific second-level views

Context	Mini-grid	Cold room
Performance	Service Dispatch Fuel Renewable Backup	Temperature Cooling Backup
Health	Asset-condition summary; click Battery / Genset / PV-Inverter / Gateway into Assets	Asset-condition summary; click Compressor / backup assets / instrumentation into Assets
Foundation	Definition Topology Controls Changes Readiness	Same structure, with refrigeration / policy / topology context

2.3 Scoped-index rule: site tabs versus the global menu

Four site tabs name things that also exist in the left navigation. They are legitimate because the site is the root object and the user is asking a site-bounded question, but they must not become second lists with their own status models. v6 splits them into two kinds:

Site tab	Kind	What the site tab owns	What stays global	Design contract
Findings	Scoped index	Nothing new: a filter over Findings plus a <i>Domain</i> column linking back to the Performance subtab that produced each finding.	Counts, severity, confidence, states, lenses, Finding Detail.	Same columns/states/lenses as the global index; no KPI strip; lens subtab row; <i>Manage in Findings</i> link; row click changes root to Findings.
Work	Scoped index	Nothing new: three filtered indexes with counts in the subtab labels.	Lifecycle states, Action / Incident / Work-request detail, create flows.	Subtab row mirrors the WORK nav group exactly: Actions Incidents Maintenance; row click changes root to that class.
Financials	Site-owned content	The technical bridge for this site (behaviour → mechanism → consequence), assumptions and scenarios.	Non-overlapping portfolio aggregation and cross-site value drivers.	Site page keeps Sites highlighted; chip <i>Portfolio roll-up in Financials</i> →; values click back to the source finding; the global page never re-derives site bridges.
Evidence	Site-owned content	Capability claim ceilings, missing/preferred evidence, source health and instrumentation priorities for this site.	Capability coverage matrix across sites and recurring gaps.	Site page keeps Sites highlighted; chip <i>Portfolio roll-up in Evidence</i> →; global cell click drills to the site page.

Test. If a site tab shows a list of global objects, it must be possible to remove that tab entirely and lose nothing except a filter. If a site tab shows content that the global page cannot show (readiness, claim ceilings, technical bridge), it is site-owned and the global page must link down to it rather than reproduce it.

2.4 Navigation ownership and scoping rules

Current root entity / workspace	Left navigation	Contextual tabs / subviews
Portfolio	Portfolio	Portfolio command-center lenses and drilldowns
Site	Sites	Overview Foundation Health Performance Findings Work Financials Evidence
Asset	Assets	Overview Health Performance Configuration Work Evidence Warranty (P1)

Finding	Findings	Overview Evidence Alternatives Recommendation History
Action / Maintenance work request	Actions / Maintenance	Overview Plan Execution Evidence Verification History
Incident	Incidents	Overview Timeline Findings Work Evidence History
Portfolio evidence	Evidence	Portfolio readiness / gaps; not site evidence
Portfolio financials	Financials	Portfolio consequences; not site financials
Simulated site	Sites	Identical to Site; header carries site_id · SIMULATED · scenario · contributing run(s) and the Replay / Open in Simulator Lab entries
Simulator Lab (separate developer workspace; feature-gated)	(own shell: Home Sites Simulator Lab Scenarios Site Templates Library Documentation Settings)	Run: Site View Configuration Environment & Scenario Events Devices & Sensors Gateway & Ingestion Logs. Scenario: Site Events Evidence Interventions Expectations Runs. Exempt from root-entity tab rules; bridge = Open in AssetOps
Replay	(unchanged)	A mode on a simulated site, never a tab: adds a controller; all tabs render recorded Site history as of time t
Simulation runs (management)	(Simulator Lab › Runs)	Developer surface listing time-bounded runs per site_id; inspect / open in AssetOps / replay / truth / rerun

Scope rule. Clicking a finding, incident, action, work request or asset from any site tab changes the root object and therefore the left-nav highlight; the breadcrumb retains the site so back-navigation is predictable. **Maximum visible depth:** main navigation → contextual entity tab → one subtab row.

3. Product backbone and object model

Site Foundation → Operational Evidence → Finding → Recommendation → Action → Investigation / Configuration Change / Test / Work Request → Verification → Outcome → Financial / Contractual Impact → Portfolio Learning

Object	Role
Site Operational Record	Time-valid identity, topology, ratings, control policy, operating obligations, responsible parties and change history.
Evidence	Observable telemetry, events, gateway health, operational records, human records and external/business context.
Finding	A bounded analytical conclusion with evidence, consequence, confidence, alternatives, claim ceiling and lens (Opportunity / Risk / none).
Incident / Case	Groups related alarms, events and findings into one managed operational episode.
Recommendation	Suggested response tied to a finding and suspected mechanism; it is not yet managed execution.

Action	Accepted managed task with owner, due date, constraints and declared success criterion.
Work Request	Field or maintenance execution object created only when an action requires physical work; shares the managed-task lifecycle.
VerificationOutcome	Compares declared success criteria against measured post-action operation and recurrence.
Assessment	Findings + trends + positive evidence + missing evidence + operating objective.

3.1 Simulation architecture

Principle. Simulation is a source mode, not a product mode. A simulated Site is a normal Site object whose operational evidence is produced by a versioned SimulationRun rather than field integrations. All downstream ingestion, evidence, findings, incidents, actions, financials and verification use the same production contracts. Simulation provenance is visible but does not alter product semantics (“Dispatch: Needs attention”, never “Simulated needs attention”). Scenario authoring and simulator truth remain outside the operational product shell.

Object	Role
Site (extended)	Stable canonical <code>site_id</code> declared in the Site definition (Site YAML for the MVP, e.g. <code>site_id: maji-mini-grid-001</code> , <code>site_kind: mini_grid</code>); <code>mode = LIVE SIMULATED</code> ; source provenance. The same <code>site_id</code> is carried through telemetry, events/alarms, operational records, gateway health, controller history, Evidence, analytics, Findings, Incidents, read models, reports/exports and provenance. Same tabs, vocabulary and claim discipline in both modes; live field writes disabled when SIMULATED. No <code>run_id</code> on the Site: runs reference the Site, not the reverse.
ScenarioTemplate	Reusable archetype baseline (e.g. PV + Battery + Diesel mini-grid; Grid + Genset cold room). Baseline fields reuse the Site Foundation schema rather than a second config schema.
ScenarioDefinition	Versioned description of <i>what happens during a simulated interval</i> : parameters, world events, control/configuration changes, evidence-quality conditions, operational-record conditions, interventions, duration / time interval, seed and private evaluation assertions. It must not specify Findings, assessment results, severity, product conclusions, Actions or VerificationOutcomes. World events and evidence manipulation are authored separately. Site Foundation answers “what is this Site?”; ScenarioDefinition answers “what happens?” – the two never duplicate each other.
SimulationRun	Time-bounded execution record of a ScenarioDefinition against a declared Site; mutable only while Draft, immutable once Committed . Two independent status dimensions: <code>lifecycle = Draft Committed</code> ; <code>execution = Ready Running Paused Completed Failed Blocked</code> . Fields: <code>run_id</code> , <code>site_id</code> , <code>scenario_id</code> , <code>scenario_version</code> , <code>simulator_version</code> , <code>seed</code> , <code>simulated_start</code> , <code>simulated_end</code> , configuration/mapping versions, source-artifact refs, private truth reference, execution status. A run contributes source evidence to the Site's history for its interval. One Site may have many runs on non-overlapping intervals; overlapping runs are guarded (§3.3). A run never allocates or defines Site identity.

SimulationTruth	Private oracle output owned by the SimulationRun / simulator evaluation context (e.g. EXTERNAL_ADJUSTMENT -80 L). Observable evidence belongs to Site history. SimulationTruth is private to the simulator / evaluation domain: it may be surfaced anywhere inside the Simulator Lab (Site View, Site state, Devices & Sensors), by the Expectations layer, and by the admin truth-comparison overlay in AssetOps, but it never enters canonical envelopes, Evidence, analytics, normal operator UI or exports.
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Mode	Foundation	Dynamics / evidence	Use
Synthetic site	Template	Simulator	Demos, regression, product development, scenario packs
Scenario clone	Cloned from a real site	Simulator (labelled clone, not a calibrated twin)	“What if reserve were 30%” experiments beside the live site in the same portfolio
Historical replay	Real site	Recorded real inputs replayed through the pipeline; optional branch	Regression against known episodes; cautious what-if

Pipeline seam. Simulator Lab / Scenarios → ScenarioDefinition → SimulationRun → simulated world → simulated devices/controllers → simulated gateway → *canonical source envelopes* → serialization/persistence → normal AssetOps ingestion → validation → normalization / mapping / quality → Evidence → analytics / read models → Site. The simulator ends at the boundary where a real site begins; downstream code does not know whether observations came from Modbus, CAN, MQTT, file import, manual record or a simulated device, except through provenance.

Rules. (1) Authors configure the world, records, evidence visibility and interventions; they never configure findings or severities. Expected findings exist only in the Expectations layer, invisible to the pipeline. (2) Every derived claim on a simulated site retains scenario ID/version, run ID, simulation clock, simulator version, seed, interventions and mapping/config version. (3) Scenario versions and runs are first-class (v4 / seed 42 → sim-2041 for Aug 5–19; v4 / seed 77 → sim-2043 for Aug 19–Sep 3, both against site_id maji-mini-grid-001). (4) Scenario packs (mini-grid operations assurance; cold-chain assurance) generate whole demonstration portfolios of simulated Sites in one action. (5) Authoring is template-first with scenario recipes (avoidable runtime, fuel delivery mismatch, tank-sensor drift, reserve breach, curtailment, outage + START_FAIL, cooling degradation, door-open excursion, gateway offline, clock drift, maintenance effective / ineffective, configuration change with trade-off), not a blank-canvas editor.

Entry points (no new operator navigation items). Sites > + Add site ▼: Register live site / Create simulated site / Clone site into scenario (the last two open the Simulator Lab workspace). Workspace utility menu: Open Simulator Lab. Simulated-site header: Replay ► (a mode, never a tab) and Open in Simulator Lab ↗ (leaves the operator shell). From the simulator: Open in AssetOps ↗ on an eligible Committed run (Draft runs show the control disabled); Open target Site ↗ from a ScenarioDefinition. Admin roles only: Compare with simulation truth.

3.2 Simulation execution contract

§3.1 fixes the principle; this subsection fixes the four implementation details that would otherwise let an implementation keep the UI semantics while bypassing the real ingestion/evidence boundary.

Contract	Rule
1. Identity and lifecycle Site ↔ SimulationRun	A Site has a stable canonical <code>site_id</code> declared in its Site definition; for the MVP this is the Site YAML (<code>sites/maji-mini-grid.yaml</code> is a <i>representation</i> of the Site Foundation, not the definition of the Site concept). A SimulationRun is immutable once committed (draft runs may be reset before commit, §3.9) and references the <code>site_id</code> it runs against; it never allocates, renames or replaces Site identity. Reruns (new seed, new scenario version, edited interval) create new runs, not new Sites. For Scenario clone and Historical replay modes the Site records <i>source_site_id</i> of the real site it derives from. Managed-object writes on a simulated Site (acknowledgements, accepted actions, notes) are stored against that <code>site_id</code>, never against any live site. Deleting or archiving a run removes its contribution to the Site history and its descendants; it never rewrites other runs. Future backend Site registration (Register Site → backend persists Site → gateway data using that <code>site_id</code> is accepted) is expected but deferred; it is not a v6.4 P0 requirement.
2. Ingestion equivalence	The simulator emits only through the same gateway/source envelopes a real integration would produce: identical message schemas, register/point mapping IDs and mapping versions, timestamps and clock-state fields, quality flags and coverage semantics, event records, and manual/human record formats. Simulated evidence enters through the normal ingestion path and is subject to the same validation, normalization, quality scoring and claim-ceiling logic. There is no simulator-only API into derived analytics: nothing may write DispatchInterval, FuelBalanceWindow, Finding, Incident or any other derived object directly. Canonical envelopes are serialized/persisted before ingestion, and the product (Site pages, analytics, read models) must be reproducible from those persisted envelopes alone – without simulator runtime objects or SimulationTruth. Permanent in-process simulator → analytics/read-model composition is not an acceptable canonical architecture. A file-backed implementation is acceptable for the MVP; a network hop is not required: production equivalence means the same logical source contract, not production networking. If a capability cannot be simulated at the source-envelope level, it is marked unavailable for that run (evidence status None / N/A), never approximated downstream.
3. Provenance propagation	Provenance (canonical <code>site_id</code>, source mode, <code>run_id</code>, <code>scenario_id</code> + version, simulator version, seed, source/gateway, simulated clock/time, interventions, mapping/config versions) is attached at the Evidence layer and is inherited, not re-derived, by every descendant: Finding, Incident, Recommendation, Action, Work Request, VerificationOutcome, financial consequence line, portfolio roll-up contribution, report and export. A descendant with any simulated ancestor is simulated. Global indexes (Findings, Actions, Incidents, Maintenance, Evidence, Financials) show the SIM marker on such rows and offer a Live / Simulated filter; portfolio roll-ups either exclude simulated contributions by default or label the roll-up as mixed. Exports carry provenance in the file header and per row. For a Site/time window spanning several runs, each evidence item and each descendant remains traceable to its contributing run(s); <code>run_id</code> is never the parent identity of the Site. Provenance is read-only and cannot be edited, stripped or promoted to live.

4. Replay semantics	Replay at time t is an as-of view of recorded Site history, not a re-simulation and not a run-centric view. (a) Only evidence whose <code>received_at</code> $\leq t$ is visible; records that arrive late (delayed operator notes, back-filled dips, delayed or out-of-order messages) appear only from their <code>received_at</code>, not their <code>source_time</code>. (b) Derived and managed objects respect their own creation and state-change times: a Finding is absent before it formed, an Action shows the state it had at t, a verification window shows its progress at t. (c) Future records, future states and future outcomes are hidden; nothing at t may reference anything after t. (d) Configuration valid at t (Foundation version, mappings, control policy) is shown, not the current one. (e) Derived objects are served from the pipeline's own recorded outputs for the Site history, not recomputed on the fly; historical conclusions are never silently recomputed with newer algorithms (a recomputation is a new, labelled analysis version). (f) Replay operates within the selected time window; <code>SimulationRun</code> is available as provenance, filter and admin context, not as the address. (g) Replay never shows <code>SimulationTruth</code> unless the admin <code>Truth-vs-observed</code> overlay is on, and that overlay is likewise as-of t.
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Test. Given a scenario run, an engineer must be able to (i) point to the persisted source-envelope files that produced every Finding, (ii) find `site_id`, run and scenario version on every descendant object and export, (iii) run a second non-overlapping interval against the same `site_id` and see one Site history with both runs traceable, and (iv) scrub Replay to any t and see only what the product had asserted by then.

3.3 Site identity and Site history

Site history = Site + selected time window. The normal AssetOps historical question is “what happened at this Site during this interval?” The user navigates by Site and time window, never primarily by `SimulationRun`. Conceptually: Site + TimeWindow $\rightarrow$ eligible evidence / state / artifacts $\rightarrow$ Site read model. `SimulationRun` is provenance and execution context, not the product-history address.

- **Stable identity.** `site_id` is declared in the Site definition and is the join key for everything (§3.1). Run identity is orthogonal to it.
- **Eligibility by window.** The selected window determines which time-relevant material may appear across the Site experience: telemetry/evidence, service state, configuration valid during the interval, analytic windows, findings, incidents, evidence-quality state, provenance and other time-valid artifacts. All Site tabs resolve against the same selected context unless a screen explicitly declares a different scope.
- **Multiple contributing runs.** One logical simulated Site may have several time-bounded runs (e.g. sim-2041 for Aug 5–19 and sim-2043 for Aug 19–Sep 3 against maji-mini-grid-001). A window spanning both shows evidence from both, each item traceable to its run.
- **Overlap guard (unresolved, explicitly guarded).** Runs that overlap the same Site/time range are never silently combined into one canonical history. Until a branch/version-selection model is designed, the system must either prevent ambiguous overlapping canonical runs or require an explicit selection/context outside normal Site history. Implementation must not choose merge behaviour implicitly.
- **Sites index.** One row per logical `site_id`, never per run. Run management lives in the simulation workspace (8.15).

3.4 Artifact time semantics

Not every object has one timestamp. The Site/time-window query defines how each class intersects the requested window:

Class	Examples	Intersects the window when
Point-in-time	telemetry samples, events/alarms, observations, manual records	its timestamp lies inside the window (arrival time governs Replay visibility)
Interval	DispatchInterval, FuelBalanceWindow, ServiceWindow, TemperatureExcursion, CoolingRecoveryWindow	its interval overlaps the window; partial overlap is shown as partial and labelled
Time-valid configuration	Site Foundation / configuration versions, mappings, control policy	its validity range overlaps the window; the version valid at each instant is used for expected-vs-actual analysis
Lifecycle	Finding, Incident, Recommendation, Action, Work Request, VerificationOutcome	it existed at any instant of the window; its state is shown as of the window end (or as of t in Replay)

3.5 Simulator setup (Simulator Lab › Scenarios / Site Templates)

Setup lives in the Simulator Lab developer workspace (§3.9), not operator navigation. Two setup paths:

Path	Steps
A. Define a simulated Site from a template	Select Site template/archetype → configure Site Foundation (site_id, display name, site_kind, objective, topology, ratings, control policy, other supported Foundation data) using the canonical Site Foundation schema → save/use the Site definition → choose ScenarioDefinition and simulated time interval → run.
B. Use an already defined Site	Select existing Site → select ScenarioDefinition / recipe → specify simulated date/time window → specify seed / parameters / evidence conditions → run. The run contributes source evidence to that Site's simulated history.

ScenarioTemplate reuses the canonical Site Foundation schema; there is no second simulator-specific Site schema. **Site Foundation = “what is this Site?”;**

ScenarioDefinition = “what happens during this simulation interval?” Run management (8.15) lists runs by site_id with inspect / open Site / replay / truth / rerun.

3.6 Interactive simulator runtime and causality

The Simulator Lab loads and operates a simulated Site as an **interactive physical world** of components (PV, battery, generator, fuel tank, loads/feeders, gateway, meters/sensors, cold-room equipment, controllers). This is the canonical simulator experience (8.12–8.13), not a future direction. Developer controls include stop/restore gateway, deliver or siphon fuel, add/remove load, fail/restore generator, trip feeder, change reserve setting, curtail PV, open cold-room door, bias/disable sensors, introduce clock drift, create/omit operational records.

Canonical causality: developer interaction → typed world/control/evidence event → simulator runtime → physical/control state transition → device observation → gateway → canonical source envelopes → normal AssetOps ingestion → Evidence → AssetOps analytics/conclusions.

Frozen principle. “The simulator changes the world; AssetOps discovers the consequence.” Simulator controls must never directly create a Finding, set an assessment state, create/resolve an Incident, create an Action, set a VerificationOutcome or alter financial conclusions.

Expectations rule. Scenario authors never inject Findings, assessments, confidence or other conclusions into the production pipeline. A private Expectations layer may define regression assertions about what the downstream product is expected to produce (e.g. “Candidate avoidable, confidence High”; “full / partial / missing evidence → High / Medium / Indeterminate”). These assertions are **test-oracle metadata only** - never Evidence, never analytic input, never crossing the ingestion boundary - and are visibly badged as such wherever shown (8.14).

Determinism identity. Scenario version + seed alone cannot reproduce an interactive run. A run is reproducible from: Site definition + ScenarioDefinition version + seed + simulator version + initial configuration/mapping versions + the **ordered intervention log with simulation timestamps** (every injected event is appended to it as a typed event). Given that identity, world evolution and published envelopes are identical regardless of execution speed. The intervention log is part of run provenance (§3.2 rule 3).

3.7 Protocol adapters and persistence direction

Protocol. Real Sites may reach the gateway/source layer via CAN bus, Modbus RTU/TCP, MQTT, HTTP/API, serial/vendor-specific protocols, file/batch or other industrial protocols: device/controller → protocol-specific adapter → gateway/source layer → canonical source envelopes → same ingestion. Simulation: simulated device → simulator gateway adapter → the same canonical envelopes. CAN/Modbus/MQTT describe *how* observations arrive; canonical source envelopes describe *what* AssetOps receives. The current simulator may simplify transport; realistic CAN-frame simulation is not required unless a use case needs protocol behaviour.

Persistence. Standing principle: **production-compatible seams, MVP-simple infrastructure.** Development/demo persistence may use YAML, JSON, filesystem artifacts and simple in-memory/file repositories, but these are representations, not domain semantics. Domain access goes through explicit repository/storage seams so that later implementations (PostgreSQL, object storage, specialized evidence storage where justified) change nothing about Site meaning, SimulationRun meaning, Evidence semantics, analytics or the Site UI.

3.8 MVP deployment / stack note

Preferred MVP (consistent with ARCHITECT_GUIDE): Python/FastAPI backend; React/TypeScript frontend; Python deterministic simulator; modular monolith; strict source/API parsers; file-backed repositories/artifacts where sufficient; simple containerized single-host deployment; CI-enforced architecture guards. Not required for MVP unless justified: Kubernetes, microservices, Kafka, service mesh, complex distributed storage, dedicated time-series infrastructure. Technology choices are not product semantics.

3.9 Simulator Lab: developer workspace and time semantics

Product boundary. The Simulator Lab is a **virtual physical site** whose purpose is to behave like a real site and publish data through the same ingestion semantics AssetOps expects in production. It owns the simulated world, environment, controllers, devices, sensors, events, time evolution and gateway publication. AssetOps begins only after the ingestion boundary. The simulator never shows AssetOps findings, analytics, confidence scores, reconciliation results, health scores or product interpretation; Normal AssetOps ingestion and the operator experience see only published observations, never simulator truth; the admin truth-comparison overlay is an out-of-band debugging capability, not an exception to the ingestion rule. **Environment / Scenario → Simulated physical site → Controllers → Devices / Sensors → Gateway → AssetOps ingestion.** The simulator knows the truth; AssetOps does not.

Workspace shell. The Simulator Lab has its own shell and navigation (Home | Sites | Simulator Lab | Scenarios | Site Templates | Library | Documentation | Settings), separate from the operator product; “Scenario Lab” in earlier versions is the Scenarios area of this workspace. Its run header shows run, status, site, scenario, template, seed, simulation window, simulated clock, wall-clock elapsed and simulated elapsed/progress; its control row shows timestep, current execution speed, Pause / Step / Fast-forward to simulated time / Reset (draft); the **Open in AssetOps** action opens the same site_id in the normal Site shell showing only published data. Tabs: Site View | Configuration | Environment & Scenario | Events | Devices & Sensors | Gateway & Ingestion | Logs. **Simulator “Sites”** in this navigation means the site definitions available to the simulator (Site Foundation records, same canonical site_id); it is not a second rendering of the AssetOps operational Sites index, and there is no simulator-owned Site identity.

Run time concept	Definition
Simulated start / end or duration	The simulation window (hours, days or weeks); the run's simulated_start / simulated_end.
Simulation timestep	Fundamental world-evolution tick (e.g. 1 s). Independent of execution speed.
Execution mode	Real time; accelerated playback (e.g. 10×, 100×); maximum speed / batch. Changing execution speed never changes the simulated result.
Simulated clock / elapsed / progress	Drives environmental forcing, controller/EMS execution, physical evolution, device sampling, events, alarms and gateway publication. All schedules are anchored to it.
Wall-clock elapsed	Execution time only; carries no simulation meaning.

Schedule (example)	Cadence on the simulation clock
World evolution	every 1 s
EMS evaluation	every 5 s
BMS telemetry	every 10 s
Inverter telemetry	every 30 s
Fuel sensor	every 60 s
Weather station	every 5 min

Lifecycle chain. Draft → simulator produces staging envelopes → Commit → seal/persist canonical source envelopes → normal ingestion → validation → normalization → mapping → Evidence → Site history / analytics.

Runtime controls. Pause freezes simulation time (Resume continues it); Step advances one fundamental tick; *Fast-forward to simulated time* executes all intervening ticks (never skips physical evolution – the control is deliberately not called “jump”). Wall-clock elapsed is shown beside *current* speed and does not imply one constant execution speed.

Run state / control	Meaning
Draft run	The simulated gateway produces canonical source envelopes into a run-scoped staging stream; nothing has crossed the ingestion boundary. Reset (draft) discards the staging stream and execution state and restarts from the deterministic identity (§3.6).
Commit run (explicit action)	Shown in the top bar when viewing an eligible Draft run and as a row action in Runs (execution Completed or Paused, overlap guard satisfied); never a list-level command and never on a ScenarioDefinition. Commit seals and persists the run's staged canonical source envelopes and releases them through the normal AssetOps ingestion path. The run becomes immutable at commit (lifecycle = Committed; Reset unavailable). Site history is populated only through normal validation, normalization, mapping, quality and Evidence processing; Commit never writes Site history or derived objects directly.
Open in AssetOps	Enabled only for Committed runs, and useful once the committed envelopes have been accepted by ingestion. For the MVP, commit is processed synchronously so the two moments coincide; a later asynchronous ingestion may show <i>Submitted</i> until acknowledgement without adding a lifecycle state. For a Draft run the control is disabled with the note “draft envelopes are not released to ingestion until Commit”.
Gateway output label	State-sensitive: Staging (Draft) → Submitted to AssetOps (Committed) → Ingested by AssetOps (acknowledged).
Status vocabulary	Lifecycle: Draft Committed. Execution: Ready Running Paused Completed Failed Blocked. The two are never combined into one status string.
Rerun	Creates a new run_id for the same site_id from the same or an edited identity; never regenerates or rewrites a committed run's envelopes.
Replay	Consumes recorded history; publishes nothing (§3.2 rule 4). Snapshot-based navigation of completed runs is a Replay capability.

Environment and scenario. Controls and state for irradiance, ambient temperature, cloud cover, wind, demand/load profiles, scheduled faults, fuel deliveries, fuel loss, sensor drift, communications outages and maintenance events. Scenario events alter the world, device, controller, sensor or communications process – never an AssetOps conclusion (§3.6).

Devices and sensors. The Devices & Sensors panel shows physical truth beside sensor-reported values (e.g. tank 274 L true / 289 L reported) with status and next-sample time, and supports noise, bias, quantization, stale values, missing samples, device failure, communication interruption and delayed recovery.

Truth visibility invariant. Simulator truth may be displayed within the Simulator Lab wherever necessary to operate and understand the synthetic physical world – including Site View, Site state and Devices & Sensors. SimulationTruth never enters gateway envelopes, normal AssetOps evidence, analytics, operator UI or exports; inside AssetOps it is readable only through the admin Truth-vs-observed overlay (8.17). The boundary is Simulator vs AssetOps, not one simulator panel.

Gateway and ingestion semantics. The Gateway & Ingestion panel shows exactly what AssetOps receives, in two layers. The *summary table* shows source time, gateway time, device, point, value, unit and quality. *Inspect raw gateway output* exposes the complete canonical envelope: site/gateway identity, device or source identity, signal/register identity and mapping id/version, value, unit, the three timestamps, sequence/message id, quality, communication state, sampling cadence, clock state, provenance/source mode, and event or alarm type where applicable. **Timestamp vocabulary (frozen):** `source_time` (device/source), `gateway_time` / `published_at` (gateway publish, where supported), `received_at` (AssetOps ingress). The simulator can display `source_time` and `gateway_time` before hand-off; `received_at` is recorded by the ingestion boundary and becomes visible (in *Inspect raw* and in AssetOps) only after receipt – the Simulator Lab never manufactures an ingress timestamp. Replay visibility is governed by `received_at`. It can reproduce missing samples, stale data, delayed, duplicate and out-of-order messages, communication gaps, sensor bias, timestamp drift and recovery after outage. *Inspect raw gateway output* exposes the exact envelope published into ingestion. The event timeline and logs describe world events, not interpretations.

3.10 Simulator Lab feature gating

Rule. The Simulator Lab is an optional, gated capability: `simulator_lab.enabled`. For the current MVP the flag is deployment-wide (sitewide); the seam is designed so the same flag becomes per-tenant later without changing product semantics.

When the flag is	Behaviour
Enabled	Simulator Lab workspace, Scenarios, Site Templates and Runs are reachable; entry points appear (Sites › + Add site › Create simulated site / Clone site into scenario; workspace utility › Open Simulator Lab; simulated-site header › Open in Simulator Lab; simulator › Open in AssetOps); admin roles may use Compare with simulation truth.
Disabled	No Simulator Lab routes, navigation, entry points or truth overlay are rendered or served (not merely hidden). Existing simulated Sites and their persisted history remain normal Sites with SIMULATED provenance; Replay stays available because it is an as-of view of recorded Site history, not a simulator function. No operator route, tab, object schema or analytic changes. Runs cannot be started, inspected or rerun.
Scope	Deployment-wide now; per-tenant later. The flag gates surfaces and execution only; it never alters evidence, findings, provenance or claim ceilings.

4. Common screen grammar and shell

Layer	Required behaviour
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Shell top bar	Portfolio/workspace switcher, global search, selected time window , user. The time window is a controlling page context: changing it changes every time-dependent tab and read model consistently according to each artifact's declared time semantics (§3.4); no screen may silently operate on an incompatible interval unless it declares its own scope (e.g. a selected-day chart inside the window). A WORKSPACE SIMULATED badge appears only when the whole workspace is simulated.
Source-mode / provenance strip	On every Site: canonical site_id and mode. On a simulated Site and every object rooted in one: SIMULATED · scenario · contributing run(s) in the selected window · (as-of time in Replay) plus the Replay and Open in Simulator Lab entries. SIMULATED is neutral provenance, not an assessment. These are the only simulation-specific additions to the shell.
Breadcrumb + title	Breadcrumb shows root object class › root object › tab › subtab; the title is the root object; the subtitle is its identity line.
Tabs / subtabs	Canonical tab set for the root object; at most one subtab row (archetype domains, lenses, or local subviews).
Chip row	Bounded assessment, confidence, claim ceiling, scope notice and cross-links. Status is separate from confidence.
Managed-object header	Action, Incident and Maintenance work-request detail open with the same five fields: Owner Due / SLA State Source Success / closure criterion.
KPI strip	3–5 cards for the selected window. Assessment cards are coloured and read Healthy / Watch / Needs attention / Degraded / Unknown; metric cards are neutral and state their basis. Scoped-index tabs have no KPI strip.
Primary analysis	Chart, event chain, reconciliation, comparable-window analysis or condition evidence, with the derived object named in the chip row.
Explanation	Why flagged, alternatives/confounders, missing evidence and next action; each panel ends with a note stating the claim boundary or the navigation it leads to.
Evidence	Coverage, quality, configuration-at-time, mapping/algorithm version and drawer drilldown when material.

5. Single status vocabulary

Dimension	Allowed values	Applies to
Assessment	Healthy Watch Needs attention Degraded Unknown	Sites, assets, performance domains, portfolio roll-ups. Unknown is never rendered as Healthy.
Confidence	High Medium Low	Findings, assessments, financial lines.
Finding severity	Critical High Medium Low	Findings; drives ranking, not assessment.
Finding state	Open Accepted Awaiting verification Resolved Reopened Dismissed	Findings (global and scoped indexes).
Managed-task state	Open In progress Due Overdue Completed Verification window Verified Ineffective Inconclusive Reopened	Actions and Maintenance work requests.

Incident state	New Acknowledged Investigating Work open Awaiting evidence Closed verified	Incidents.
Evidence status	Full Limited None N/A	Capability coverage (site and portfolio Evidence).
Basis	Measured Modelled Extrapolated Scenario Derived Observed absence Unknown	Every number, financial value and evidence row.
Dispatch interval class	Necessary Candidate avoidable Indeterminate	Dispatch and Renewable views.
Source health	Online Stale Offline + quality grade (Good / Degraded / Bad)	Gateways, meters, sensors, connectors. Never uses the assessment vocabulary.
Colour	Green = Healthy/pass; amber = Watch; orange = Needs attention; red = Degraded/fail; grey = Unknown/indeterminate; purple = scenario/claim boundary/simulated; teal = informational	Presentation only; never the semantic status.

Scope of the assessment vocabulary. Healthy / Watch / Needs attention / Degraded / Unknown applies to the overall assessment of a site, asset, performance domain or portfolio roll-up. Sub-indicator cards inside a screen (e.g. Thermal exposure: Normal, Charge acceptance: Stable, Usable capacity: Uncertain) are local condition descriptors chosen by the domain algorithm; they are not an alternate status enum, must not be persisted as one, and are labelled as descriptors on the screen. Financial pages carry no assessment at all.

6. Claim and financial discipline

- **Fuel:** display unexplained residual / variance with uncertainty; never infer theft from reconciliation alone.
- **Cold room:** display temperature excursion and potential exposure; never convert this directly into a spoilage determination.
- **Battery:** display observable stress/condition evidence; do not claim precise remaining life without a validated field model and sufficient history.
- **Dispatch:** classify intervals as Necessary, Candidate avoidable or Indeterminate unless stronger evidence justifies an upgraded claim.
- **Maintenance:** work completion is activity evidence; verified resolution requires measured outcome against a declared criterion.
- **Financials:** mark values as measured, modelled, extrapolated or scenario-based; use ranges and prevent double counting; scoped and global pages never re-derive each other's numbers.

7. Canonical screen inventory

Scope	Screens (section)
Global collections	Portfolio command center (8.1); Sites (8.2); Assets (8.3); Findings with lenses (8.4); Actions (8.5); Incidents (8.6); Maintenance (8.7); Evidence (8.8); Financials (8.9)

Foundation / lifecycle	Site foundation and readiness (8.10); Configuration change and effect (8.11)
Simulator Lab (separate workspace)	Site View (8.12); Devices & Sensors / Gateway & Ingestion (8.13); Scenarios authoring (8.14); Runs (8.15)
Simulation in the operator product	Simulated site in Replay mode (8.16); Truth vs observed, admin only (8.17)
Mini-grid site	Overview (8.18); Health (8.19); Performance: Service / Dispatch / Fuel / Renewable / Backup (8.20–8.24); Findings scoped index (8.25); Work scoped index (8.26); Evidence (8.35); Financials (8.36)
Asset	Battery asset Health (8.27) as the implemented example of the generic Asset Detail shell
Cold-room site	Overview (8.28); Performance: Temperature / Cooling / Backup (8.29–8.31)
Cross-cutting detail	Finding detail (8.32); Incident / case (8.33); Maintenance work detail – Verification (8.34); Claim evidence drawer (8.37); Action verification (8.38)

MVP emphasis. P0: stable declarative site_id carried end-to-end, Site + time-window history, simulated Sites as normal Sites with provenance, Simulator Lab with Site View, Devices & Sensors and Gateway & Ingestion panels, simulation-time controls, both setup paths and events + evidence conditions, file-backed canonical envelopes through normal ingestion, site foundation/readiness, portfolio command center, Sites/Findings/Actions queues, both site overviews, scoped Findings/Work tabs, claim evidence drawer, incident/work/verification, and the four strongest finding pipelines (avoidable runtime, fuel reconciliation, backup failure, cooling degradation). P1: global Assets, Evidence, Financials and Reports, provided the site-scoped versions ship first. Deferred (not P0): backend Site registration, snapshot-based navigation of completed runs, branch/version selection for overlapping runs, real protocol adapters.

8. Canonical visual screens

The following mockups are the implementation baseline. They are generated from one shell and one visual grammar so that the same element means the same thing on every screen.

8.1 Portfolio command center

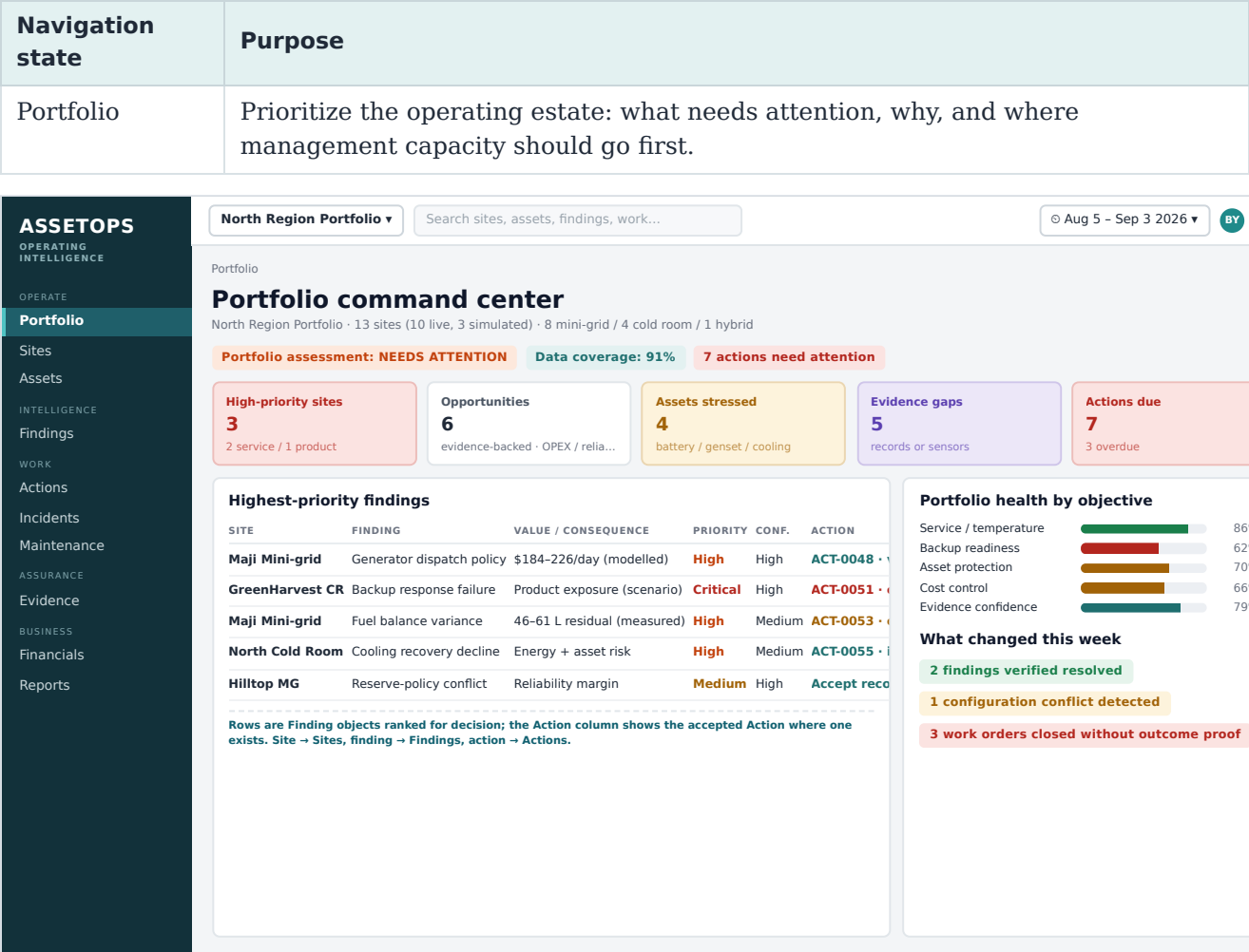


Figure 1. Portfolio command center.

Content and evidence behaviour. Uses site assessments, evidence-backed opportunities, asset-condition summaries, evidence gaps and due actions. Portfolio assessment is a deterministic roll-up; colour is presentation, not the semantic status. Value cells always carry their basis (measured / modelled / scenario).

Primary interaction. Click a site to Sites, a finding to Findings, or an action to Actions. The ranked list is a lens over Findings, not a separate queue; the Action column is the only Action reference on the page.

8.2 Sites index

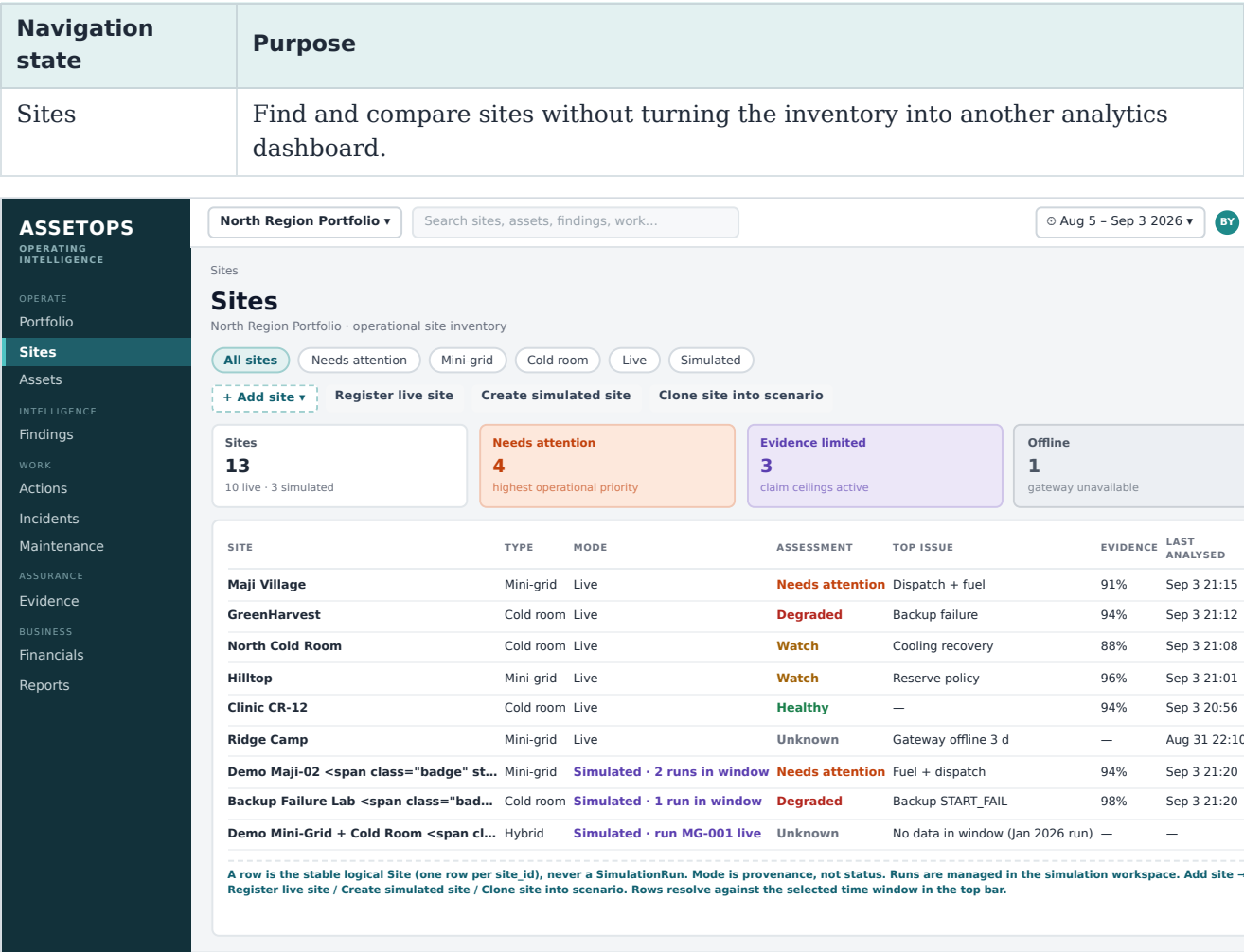


Figure 2. Sites index.

Content and evidence behaviour. Shows archetype, mode (Live / Simulated + number of contributing runs in the selected window), canonical assessment, top issue, evidence readiness and last analysis time. One row per canonical site_id; a simulated Site may have several time-bounded runs behind its history. A site with no data in the window is Unknown, never Healthy.

Primary interaction. Clicking a row opens that Site with Sites still highlighted.

8.3 Assets index

Navigation state

Purpose

Assets

Search and prioritize physical assets across sites.

ASSETOPS

OPERATING INTELLIGENCE

OPERATE

Portfolio

Sites

Assets

INTELLIGENCE

Findings

WORK

Actions

Incidents

Maintenance

ASSURANCE

Evidence

BUSINESS

Financials

Reports

North Region Portfolio ▾

Search sites, assets, findings, work...

Aug 5 - Sep 3 2026 ▾

BY

Assets

Assets

Cross-site asset inventory · condition and work context

All assets

Needs attention

Work due

Evidence limited

Assets

64

registered

Needs attention

7

condition or evidence

Work due

5

next 30 days

Evidence limited

6

asset claims constrained

ASSET	TYPE	SITE	ASSESSMENT	EVIDENCE	OPEN FINDINGS	OPEN WORK
BAT-001	Battery	Maji Village	Watch	Medium-high	2	0
GEN-001	Genset	Maji Village	Healthy	High	1	1
INV-001	PV inverter	Maji Village	Healthy	High	0	0
CMP-003	Compressor	North Cold Room	Needs attention	Medium	2	1
ATS-007	ATS	GreenHarvest	Degraded	High	1	1
GW-004	Gateway	Ridge Camp	Unknown	None	0	1

Asset condition (assessment) and evidence quality are separate columns and never merged.

Figure 3. Assets index.

Content and evidence behaviour. Shows asset identity/type, parent site, condition assessment, evidence quality, open findings and open work. Asset condition is separate from data quality.

Primary interaction. Clicking an asset opens its Asset Detail tabs with Assets highlighted.

8.4 Findings index (with Opportunities and Risks lenses)

ASSETOPS
OPERATING
INTELLIGENCE

OPERATE

Portfolio

Sites

Assets

INTELLIGENCE

Findings

WORK

Actions

Incidents

Maintenance

ASSURANCE

Evidence

BUSINESS

Financials

Reports

North Region Portfolio

Search sites, assets, findings, work...

Aug 5 - Sep 3 2026

BY

Findings

Canonical analytical conclusions across the portfolio

All findings Opportunities Risks

Lenses = interpretation (row above) Filters = state: Open Confidence ≥ Medium All sites

Open findings
18
portfolio-wide

Critical / High
6
needs decision

Low confidence
3
evidence constrained

Awaiting verification
4
action taken

FINDING	SITE	SEVERITY	CONFIDENCE	STATE	PRIMARY CONSEQUENCE	LENS
Generator support overlap	Maji Village	High	High	Open	\$184-226/day candidate	Opportunity
Backup response failure	GreenHarvest	Critical	High	Open	Product exposure	Risk
Fuel balance variance	Maji Village	High	Medium	Open	46-61 L residual	Risk
Cooling recovery degradation	North Cold Room	High	Medium	Accepted	Energy + reliability	Risk
Battery reserve stress	Hilltop	Medium	High	Awaiting verification	Reliability margin	Risk
PV curtailment displacement	Maji Village	Medium	Medium	Open	48-63 kWh/day	Opportunity

Lens is a classification of a finding, not a second object. Workflow state (Open → Accepted → Awaiting verification → Resolved / Reopened / Dismissed) is a filter chip, not a lens.

Figure 4. Findings index (with Opportunities and Risks lenses).

Content and evidence behaviour. Findings carry severity, confidence, workflow state, bounded consequence and a lens (Opportunity / Risk / none). Opportunities and Risks were separate left-nav items in v5; because they are decision lenses over findings rather than object classes, v6 makes them the subtab row of Findings. v6.1 keeps the subtab row to interpretation lenses only; workflow state is a filter chip so lenses and states are never mixed on one dimension.

Primary interaction. Clicking a finding opens Finding Detail.

8.5 Actions index

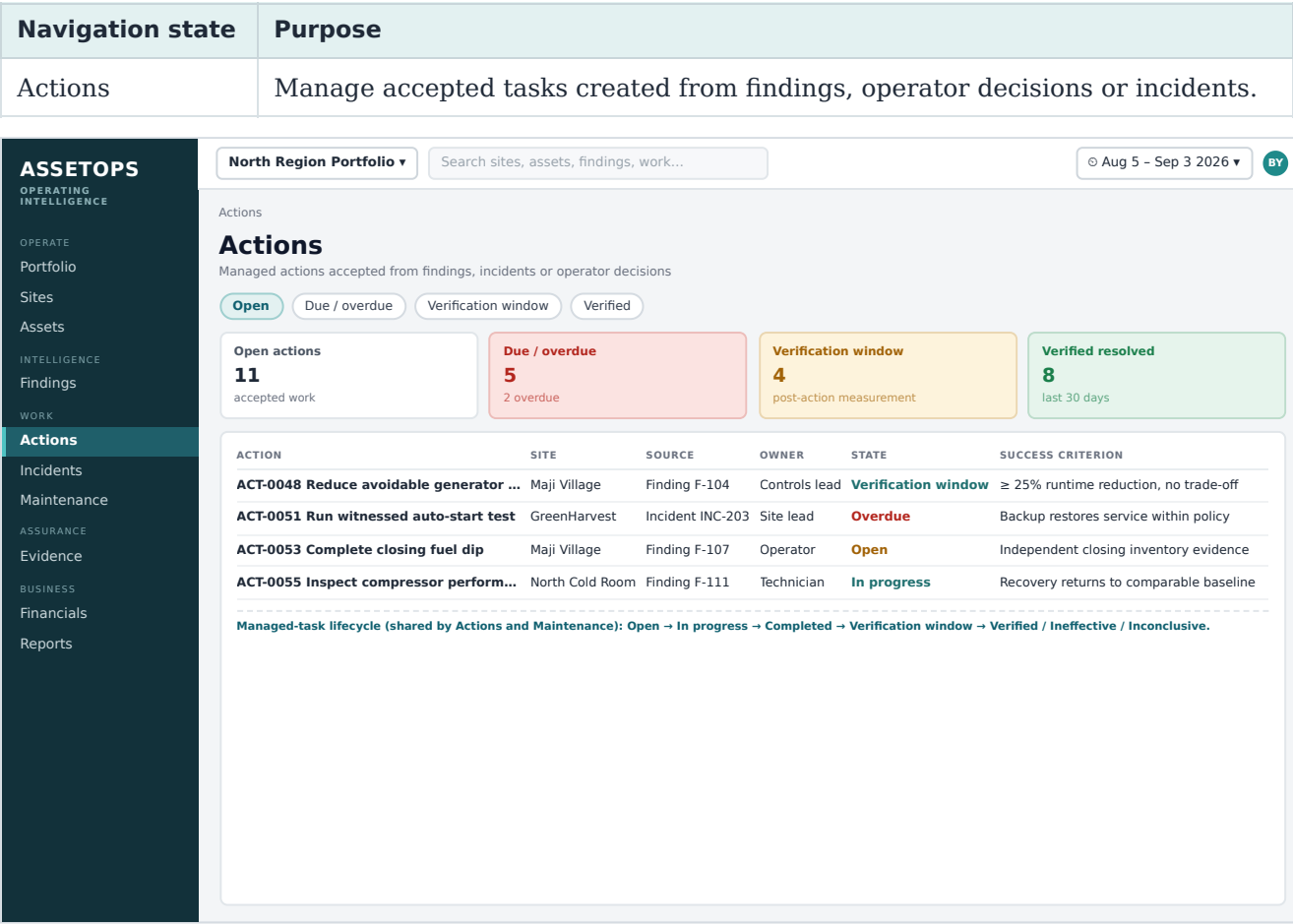


Figure 5. Actions index.

Content and evidence behaviour. An Action has owner, source, state and a declared success criterion. It may be fulfilled by a test, configuration change, investigation or maintenance work request.

Primary interaction. Clicking an action opens Action Detail; verification is part of the action lifecycle.

8.6 Incidents index

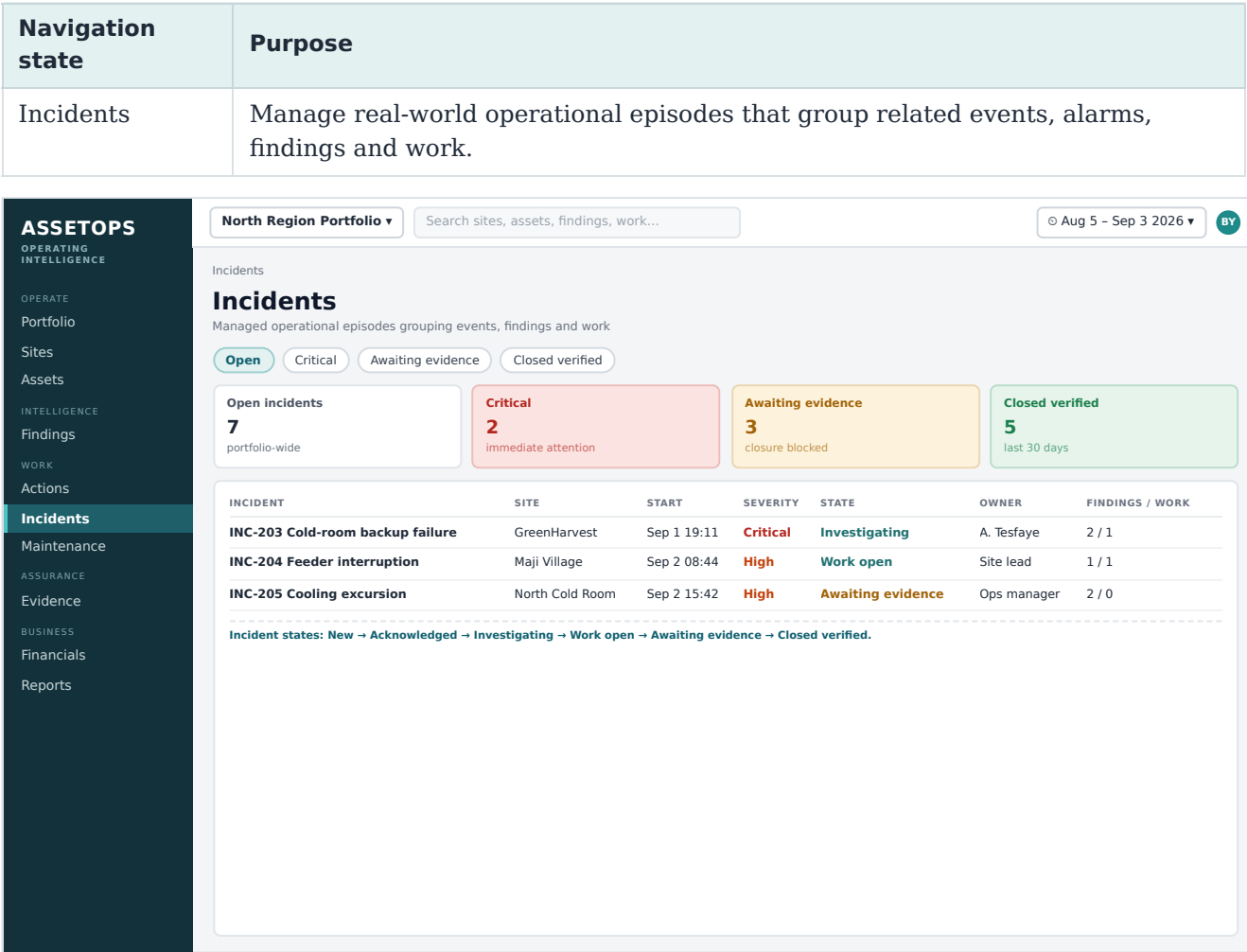


Figure 6. Incidents index.

Content and evidence behaviour. An Incident does not replace source events or findings; it provides acknowledgement, ownership, escalation and closure context.

Primary interaction. Clicking an incident opens its ordered timeline and linked findings/work.

8.7 Maintenance index

Navigation state

Purpose

Maintenance

Plan and assure field maintenance across sites without becoming a full CMMS.

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Search sites, assets, findings, work...

Aug 5 - Sep 3 2026

BY

Maintenance

Cross-site maintenance work requests and assurance

Open work

Verification due

Preventive

Calibration

Open work

14

field + workshop

Overdue verification

3

outcome not proven

Due next 30 d

8

planned maintenance

Repeat findings

2

effectiveness concern

WORK REQUEST	ASSET / SITE	TYPE	OWNER	STATE	VERIFICATION
WR-0192 Replace starter relay	CMP-003 / GreenHarvest	Corrective	Tech A	Completed	Pending - witness test missing
WR-0195 ATS diagnostic	ATS-007 / GreenHarvest	Diagnostic	Tech B	In progress	Required
PM-0441 Genset service	GEN-001 / Maji Village	Preventive	Tech C	Due	Witness test
WR-0201 Tank sensor calibration	Tank / Maji Village	Calibration	Ops lead	Open	Calibration record

Completed is activity evidence only. A work request reaches Verified only through the shared Verification tab.

Figure 7. Maintenance index.

Content and evidence behaviour. Shows open/overdue work, preventive items and verification state. Completed work is not equivalent to resolved findings. Work requests use the same managed-task lifecycle and tab set as Actions.

Primary interaction. Clicking a work item opens Maintenance Work Detail.

8.8 Portfolio evidence

Navigation state	Purpose
Evidence	Show where the portfolio has strong evidence and where claim ceilings are limiting decisions.

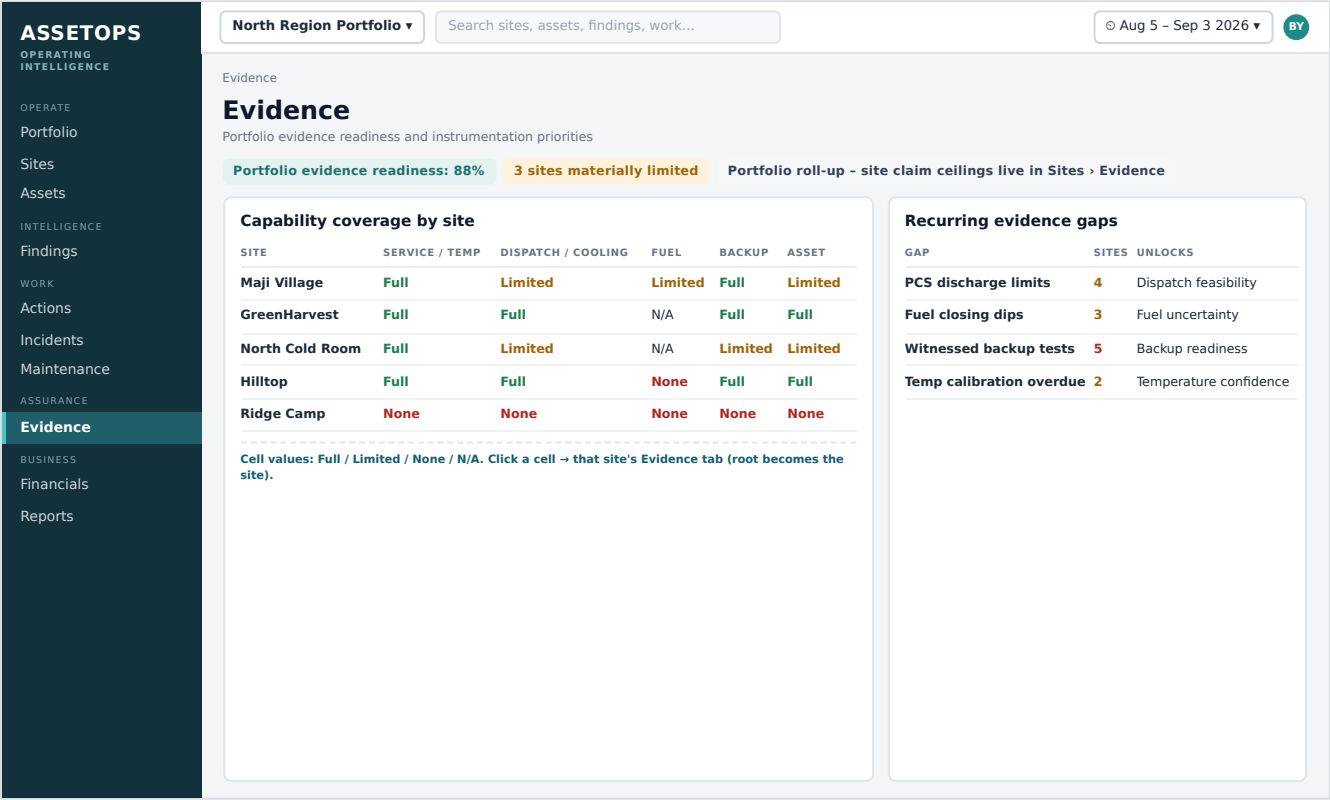


Figure 8. Portfolio evidence.

Content and evidence behaviour. Rolls up capability coverage by site and recurring instrumentation/record gaps. It is distinct from Site Evidence (per-site claim ceilings) and the claim-specific Evidence Drawer.

Primary interaction. Click a site/capability to Site Evidence; the left-nav highlight moves to Sites.

8.9 Portfolio financials

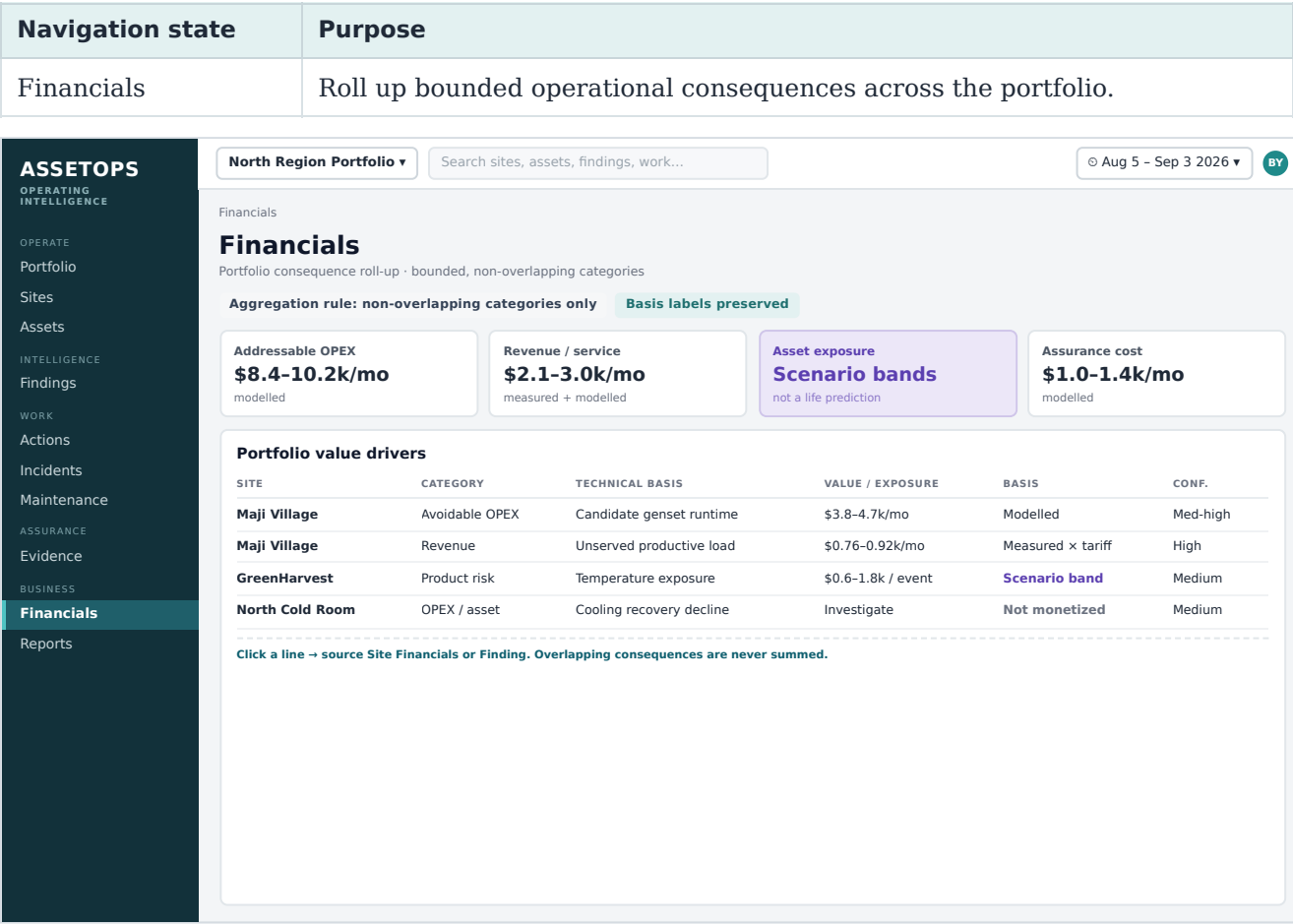


Figure 9. Portfolio financials.

Content and evidence behaviour. Only non-overlapping categories aggregate. Values retain basis labels (measured / modelled / extrapolated / scenario); asset exposure is not a remaining-life prediction.

Primary interaction. Click a line to the source Site Financials (root becomes the site) or Finding.

8.10 Site foundation and readiness

Navigation state	Purpose
Sites → Site → Foundation → Readiness	Define what the site is, what it must achieve, and whether evidence is sufficient to evaluate it.

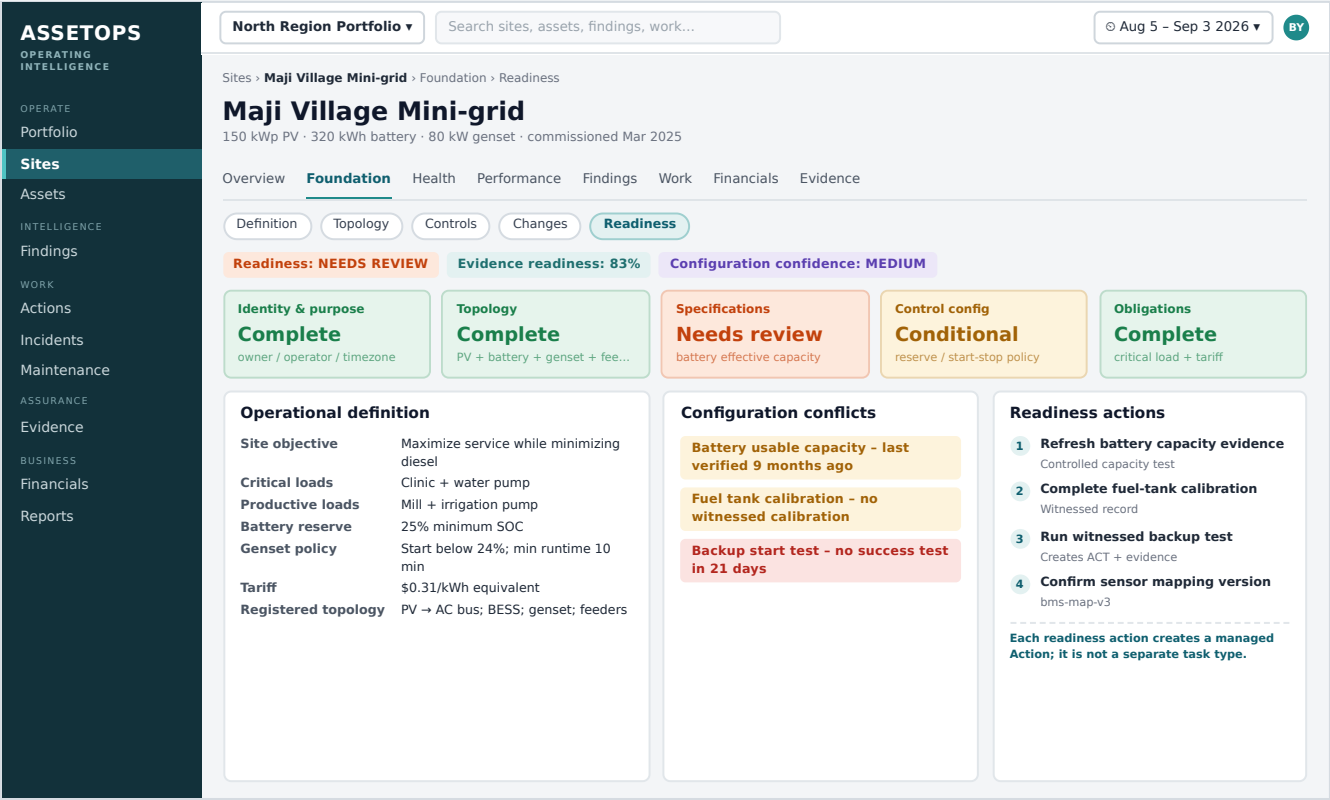


Figure 10. Site foundation and readiness.

Content and evidence behaviour. Contains identity/purpose, topology, specifications, control configuration, operating obligations and evidence readiness. Foundation uses the one allowed subtab row (Definition | Topology | Controls | Changes | Readiness). Time-valid configuration is the baseline for expected-versus-actual analysis.

Primary interaction. Readiness actions create Actions (Work) that lead to commissioning evidence, configuration review or instrumentation work.

8.11 Configuration change and effect

Navigation state	Purpose
Sites → Site → Foundation → Changes	Treat a configuration change as an auditable intervention and measure its trade-offs.

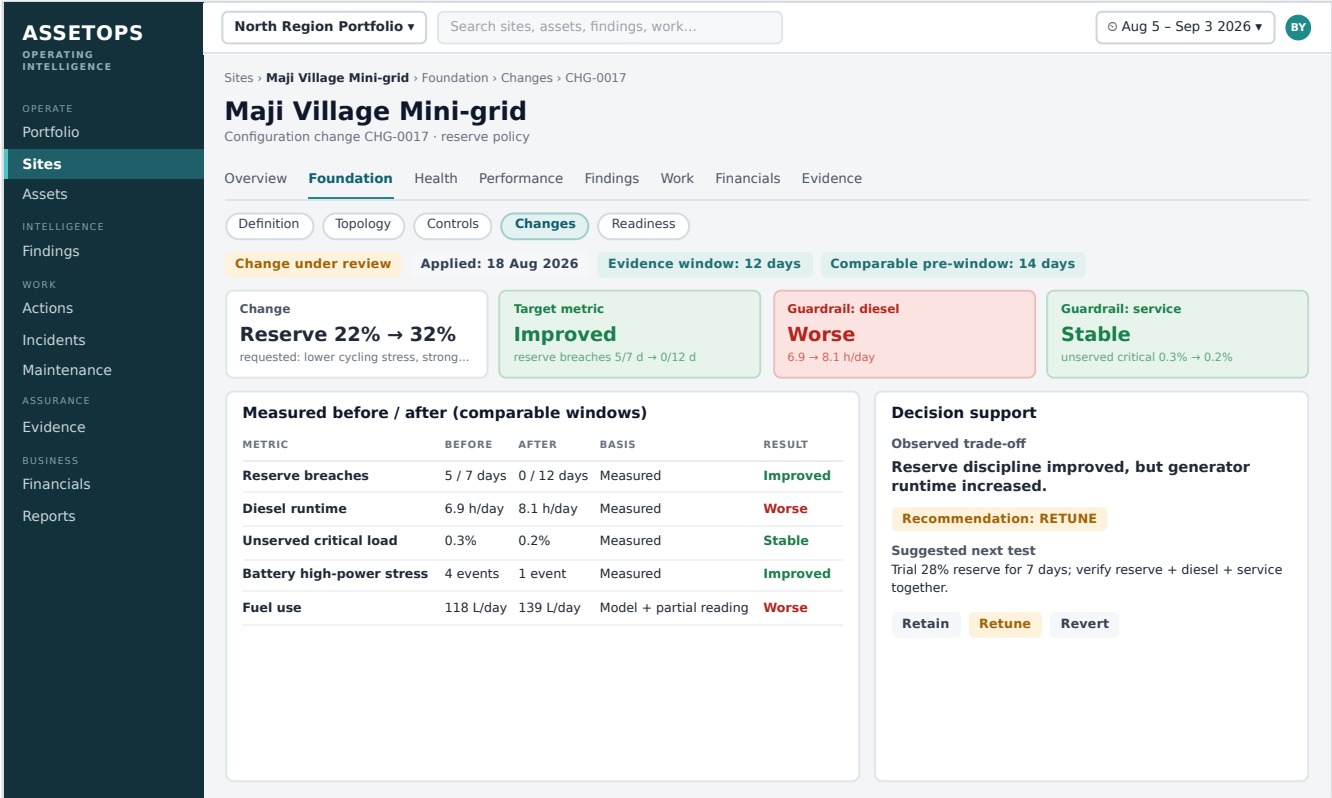


Figure 11. Configuration change and effect.

Content and evidence behaviour. Compares comparable pre/post windows across the intended target and guardrail metrics; one improved KPI is not enough to declare success. KPI cards separate target from guardrails so the trade-off is visible without the table.

Primary interaction. Retain, retune or revert based on measured multi-objective outcome; a retune creates a new change record and Action.

8.12 Simulator Lab - Site View (single line diagram, live simulation)

Navigation state	Purpose
Simulator Lab workspace → Runs → MG-001 → Site View (separate developer shell; not operator navigation)	Operate a virtual physical site - environment, controllers, devices, sensors, events and time - that publishes through the same ingestion semantics as a real site. Shows simulator truth; never AssetOps conclusions.

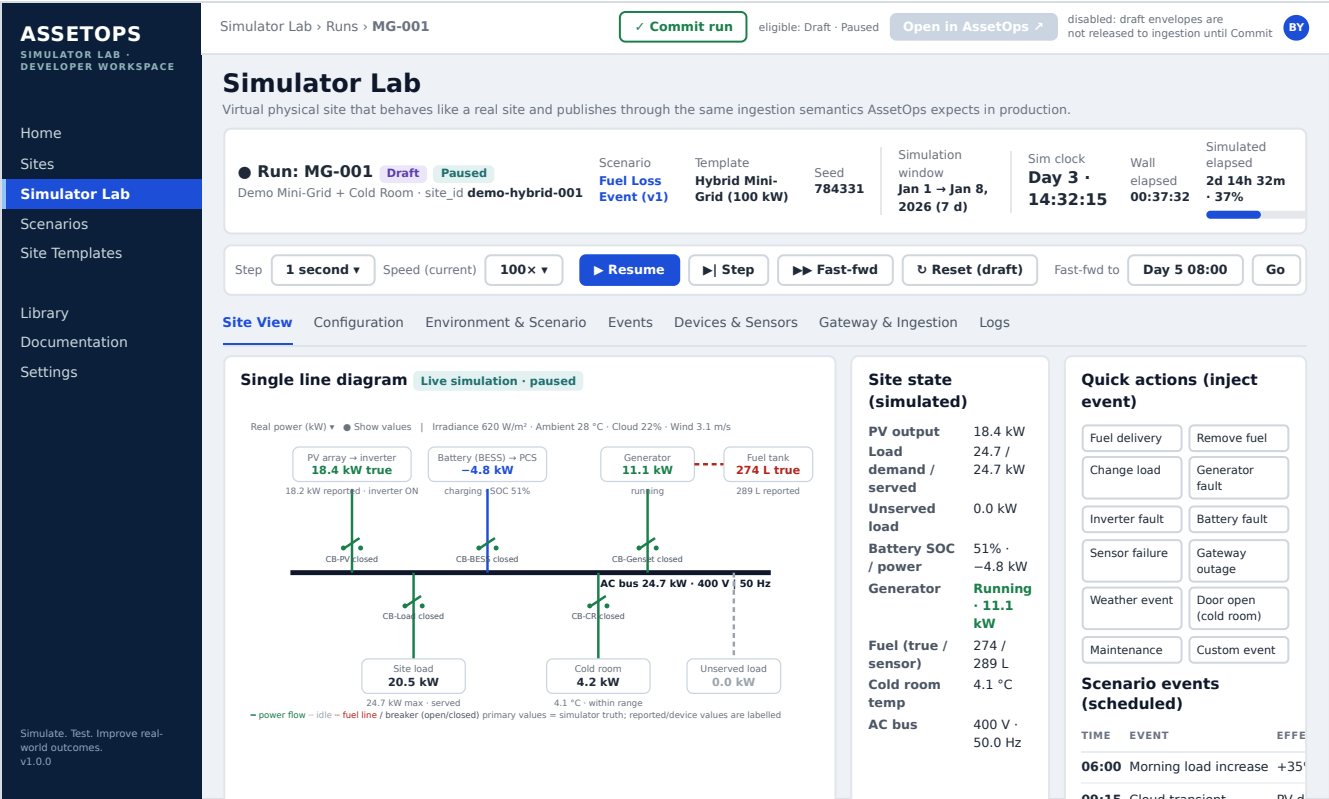


Figure 12. Simulator Lab - Site View (single line diagram, live simulation).

Content and evidence behaviour. The Simulator Lab is a separate developer workspace with its own shell and navigation (Home | Sites | Simulator Lab | Scenarios | Site Templates | Library | Documentation | Settings); it is not part of operator navigation and is gated by the `simulator_lab` feature flag (\$3.10). The Site View is a single line diagram: sources (PV → inverter, battery → PCS, generator + fuel tank) and loads (site load, cold room, unserved load) connected to the AC bus through breakers (CB-PV, CB-BESS, CB-Genset, CB-Load, CB-CR) with bus voltage/frequency, a measurement selector (real power, reactive power, current, voltage) and a show-values toggle. The run header carries run, status, scenario, template, seed, simulation window, simulated clock, wall-clock elapsed and simulated elapsed/progress; the control row carries timestep, current execution speed, Resume|Pause / Step / Fast-forward to simulated time (shortened to “Fast-fwd” in the dense UI; the accessible name is canonical) / Reset (draft). The canonical example run is Draft · Paused · 37%, which is why Commit run is eligible in the top bar. Example figures are consistent: 2d 14h 32m = 37% of 168 h, and 00:37:32 wall time at 100x; speed may change during a run, so wall elapsed never implies one constant speed. The workspace owns the simulated world, environment, controllers, devices, sensors, events, time evolution and gateway publication, and displays physical truth. It never shows findings, analytics, confidence, reconciliation results, health scores or product interpretation.

Primary interaction. Quick actions inject typed events (appended to the intervention log); the schematic and site state update on the simulation clock. Commit run (eligible Drafts only) seals the staged envelopes and releases them through normal ingestion; Open in AssetOps is disabled until then and afterwards opens the same site_id in the normal Site shell showing only ingested data.

8.13 Simulator Lab - Devices & Sensors, Gateway & Ingestion, Event timeline

Navigation state	Purpose
Simulator Lab → Runs → MG-001 → Devices & Sensors / Gateway & Ingestion	Show the distinction between physical truth and sensor-reported values, and inspect the exact canonical source envelopes produced at the gateway boundary - what will cross (Staging) or has crossed (Submitted / Ingested) into normal AssetOps ingestion.

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Simulator Lab > Runs > MG-001

Commit run

eligible: Draft · Paused

Open in AssetOps

disabled: draft envelopes are not released to ingestion until Commit

BY

Simulator Lab

Virtual physical site that behaves like a real site and publishes through the same ingestion semantics AssetOps expects in production.

Run: MG-001

Draft

Paused

Scenario

Fuel Loss Event (v1)

Template

Hybrid Mini-Grid (100 kW)

Seed

784331

Simulation window

Jan 1 → Jan 8, 2026 (7 d)

Sim clock

Day 3 · 14:32:15

Wall elapsed

00:37:32

Simulated elapsed

2d 14h 32m · 37%

Step

1 second

Speed (current)

100x

Resume

Step

Fast-fwd

Reset (draft)

Fast-fwd to

Day 5 08:00

Go

Site View

Configuration

Environment & Scenario

Events

Devices & Sensors

Gateway & Ingestion

Logs

Devices & sensors (live) - truth vs reported

DEVICE	SIGNAL	TRUTH	REPORT
PV inverter	AC power	18.4	18.2
BMS	SOC	51.2	50.5
Fuel tank sensor	Level	274	289 (
Generator ECU	Output	11.1	11.0
Site meter	Load	24.7	24.6
Cold room sensor	Temp	4.1	4.0
Weather station	Irradiance	620	618

All devices Online. Device realism: noise, bias, quantization, stale/missing samples, failure, comms interruption, delayed recovery. Truth never enters the gateway envelopes.

Gateway output - STAGING

Inspect raw gateway output

SOURCE TIME	GATEWAY TIME	DEVICE	POINT	VALUE	UNIT	QUALITY
14:32:12	14:32:13	bms-001	soc	50.5	%	Good
14:32:10	14:32:13	pv-001	ac_power	18.2	kW	Good
14:32:10	14:32:13	gen-001	output_power	11.0	kW	Good
14:31:55	14:32:13	tank-001	level	289	L	Good
14:32:10	14:32:13	meter-001	load	24.6	kW	Good
14:32:00	14:32:13	cold-001	temperature	4.0	°C	Good
14:30:00	14:32:13	weather-001	irradiance	618	W/m²	Good

Gateway: Online site_id demo-hybrid-001 · gw-001 · source_mode SIMULATED · 1,248 staged envelopes · last 14:32:13

Destination: AssetOps canonical ingestion · release on Commit · not yet received by AssetOps

Summary columns only. Inspect raw exposes the full envelope: site_id, gateway id, mapping id/version, source_time, gateway_time, sequence/message id,

Event timeline (simulation)

14:32

Inverter setpoint adj.

14:31

Tank level dropped

14:31

Gateway published

14:30

EMS reserve check

14:20

Cloud transient ended

14:00

Fuel removed -40 L

11:40

Generator started

Figure 13. Simulator Lab - Devices & Sensors, Gateway & Ingestion, Event timeline.

Content and evidence behaviour. Devices & Sensors shows physical truth next to sensor-reported values with device status and next-sample time. Gateway & Ingestion shows the canonical source envelopes produced by the simulated gateway and is state-sensitive: Staging while the run is Draft (produced, not yet released to ingestion), Submitted to AssetOps after Commit, Ingested by AssetOps once ingestion has acknowledged receipt. It reproduces missing, stale, delayed, duplicate and out-of-order messages, comms gaps, sensor bias, timestamp drift and recovery after outage. The event timeline records site/world events.

Primary interaction. Inspect raw gateway output opens the exact payload/envelope being published into ingestion. Rows link to device detail; the gateway output represented here is the only simulator-produced information that crosses into normal AssetOps ingestion; AssetOps then derives Evidence and analytics from those envelopes, never from the panel.

8.14 Simulator Lab - Scenarios (authoring: Events & evidence)

Navigation state	Purpose
Simulator Lab workspace → Scenarios → MG-HYBRID-04 → v4	Author a versioned scenario as an overlay on an archetype template, including what can and cannot be observed; produce a time-bounded run that contributes source evidence to a declared Site's history.

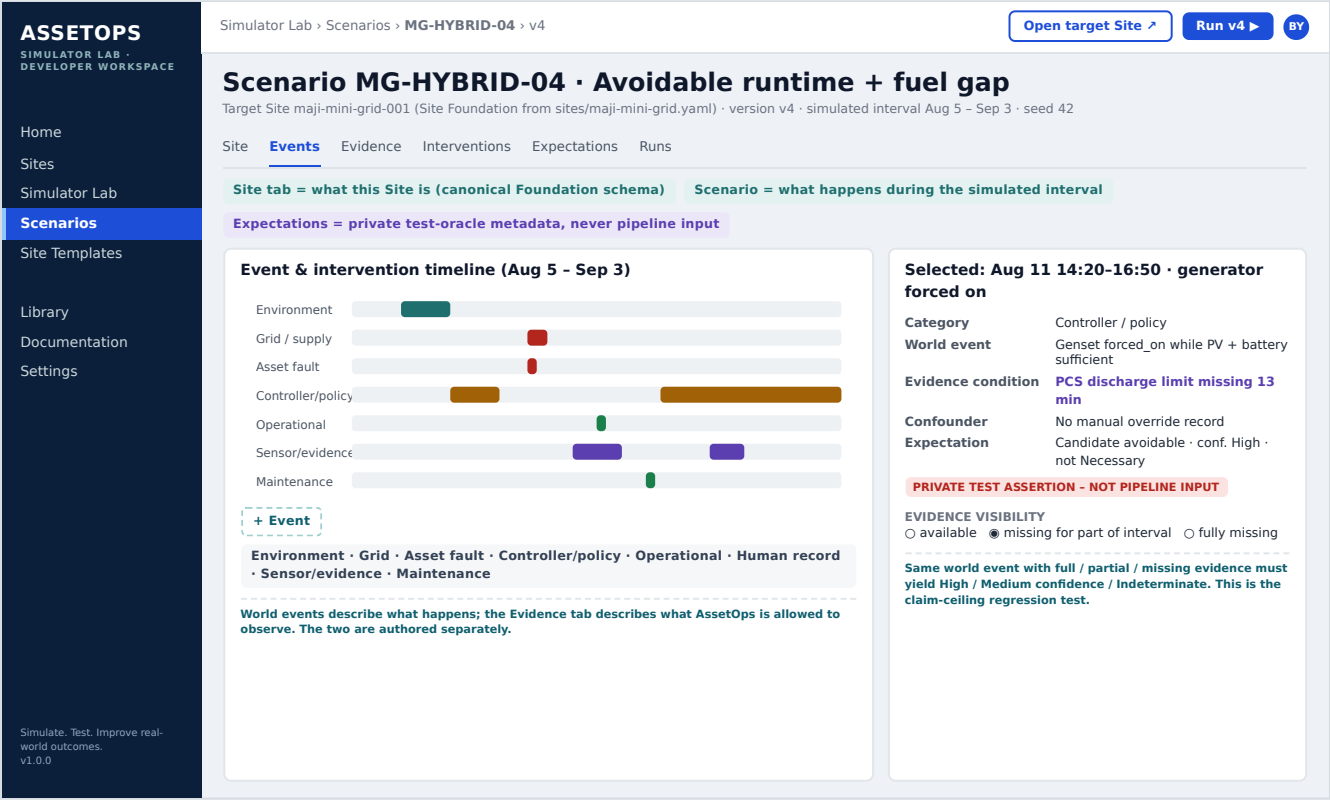


Figure 14. Simulator Lab - Scenarios (authoring: Events & evidence).

Content and evidence behaviour. Two setup paths (§3.5): (A) define a simulated Site from a template - the Site tab captures site_id, display name, site_kind, objective, topology, ratings and control policy using the canonical Site Foundation schema; or (B) select an already defined Site. Either way the ScenarioDefinition (events, control changes, evidence-quality and operational-record conditions, interventions, interval, seed, private assertions) describes what happens during the simulated interval. Authors never configure findings, assessments, severities, actions or verification outcomes.

Primary interaction. Run v4 is the page's only primary action: it creates a **Draft** SimulationRun against the declared site_id, begins execution into the run's staging stream and navigates to the run (8.12 / 8.15). Commit belongs to the run, not to the ScenarioDefinition, so no Commit action exists here; Open target Site opens the declared Site in AssetOps regardless of whether this scenario has been run. It never creates or renames a Site.

8.15 Simulator Lab - Runs (run management)

Navigation state	Purpose
Simulator Lab workspace → Runs	List and manage time-bounded SimulationRuns as execution/provenance records, separately from the Sites index.

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Simulator Lab > Runs

BY

Simulation runs

Runs are execution records against declared Sites, not Sites

All runsCompletedRunningDraftBlocked / failedBy Site

A Site row in Sites never equals a runOverlap guard: overlapping runs blocked

Reset only while Draft; committed runs are rerun under a new run_id

RUN	SITE_ID	SCENARIO / VER	SEED	INTERVAL	LIFECYCLE	EXECUTION	ACTIONS
sim-2041	maji-mini-grid-001	MG-HYBRID-04 v4	42	Aug 5 - Aug 19	Committed	Completed	Inspect · Open in AssetOps · Replay · ...
sim-2043	maji-mini-grid-001	MG-HYBRID-04 v4	77	Aug 19 - Sep 3	Committed	Completed	Inspect · Open in AssetOps · Replay · ...
sim-2044	maji-mini-grid-001	MG-HYBRID-04 v5	42	Aug 10 - Aug 14	Draft	Blocked: overlaps 2041	Needs branch context
sim-2042	cold-room-lab-001	CR-STARTFAIL-02 v1	7	Aug 30 - Sep 3	Committed	Completed	Inspect · Open in AssetOps · Replay · ...
sim-2045	cold-room-lab-001	CR-STARTFAIL-02 v1	8	Sep 3 - Sep 6	Draft	Running · 62%	Inspect · Site View
MG-001	demo-hybrid-001	Fuel Loss Event v1	784331	Jan 1 - Jan 8	Draft	Paused · 37%	Site View · Reset · Commit run

Lifecycle (Draft | Committed) and execution (Ready | Running | Paused | Completed | Failed | Blocked) are separate columns. Commit is a command on one specific Draft run: a row action for eligible drafts (execution Completed or Paused, no overlap) and a header action when viewing that run; there is no list-level Commit. Open in AssetOps appears only for Committed runs.

Figure 15. Simulator Lab - Runs (run management).

Content and evidence behaviour. SimulationRun carries run_id, site_id, scenario_id/version, simulator_version, seed, simulated_start/end, configuration/mapping versions, source-artifact refs, a private truth reference and two status dimensions (lifecycle Draft | Committed; execution Ready | Running | Paused | Completed | Failed | Blocked). A run never allocates Site identity. Two runs for one Site on non-overlapping intervals both contribute to that Site's history; a run overlapping an existing canonical run for the same Site is blocked until an explicit branch/version-selection model exists.

Primary interaction. Draft: Site View, Reset, Commit run (eligible drafts only). Committed: Inspect, Open in AssetOps (Sites highlighted, window preset), Replay, Compare truth vs observed (admin), Rerun (new run_id). Open in AssetOps is disabled for Draft runs because staged envelopes are not in canonical Site history.

8.16 Simulated site - Overview in Replay mode (Site + time window)

Navigation state	Purpose
Sites → Demo Maji-02 (simulated) → selected window → Overview → Replay	Explore a simulated Site with exactly the normal site shell. The selected time window is the controlling context; Replay is an as-of view of recorded Site history at time t.

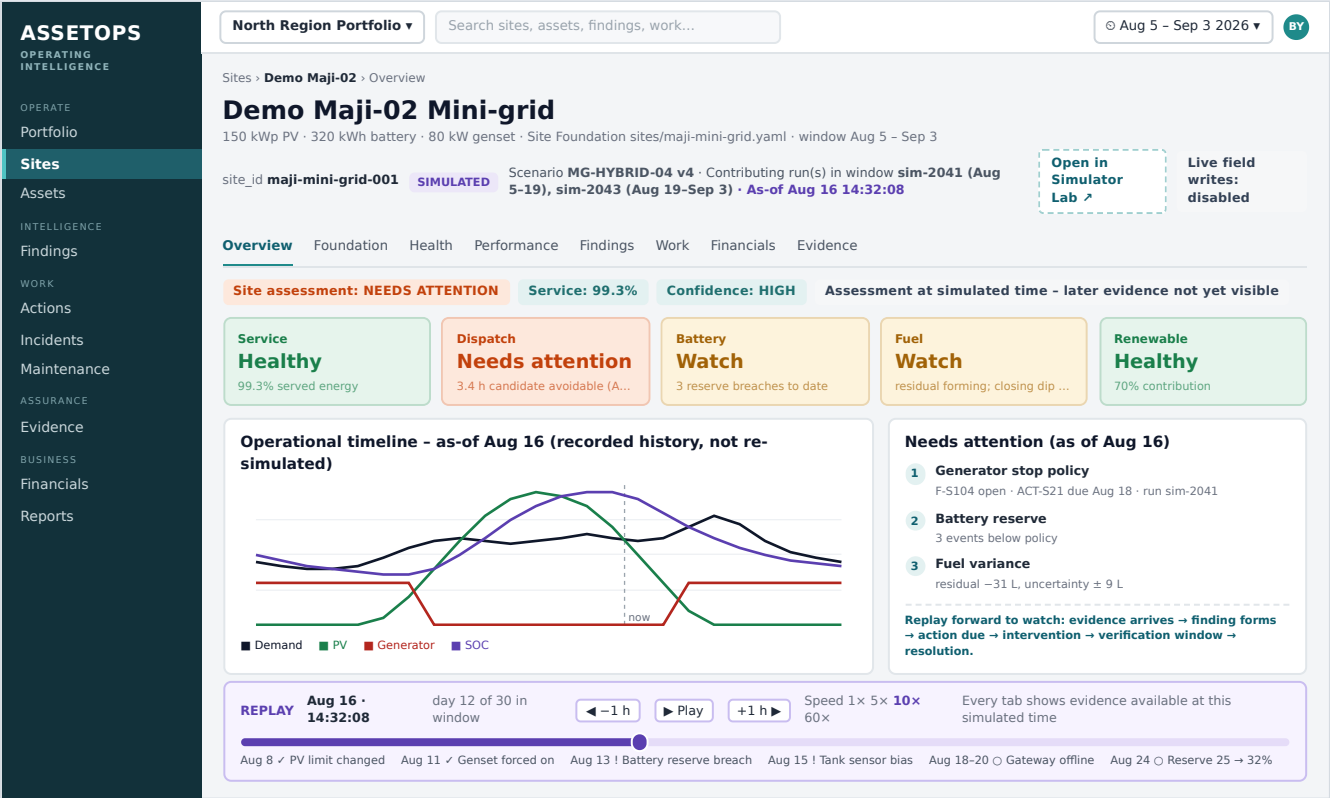


Figure 16. Simulated site - Overview in Replay mode (Site + time window).

Content and evidence behaviour. A simulated Site is a normal Site object with a stable site_id and mode = SIMULATED. Its history for the selected window is assembled from all contributing time-bounded runs (here two, non-overlapping) with run provenance preserved per evidence item. Nothing in the tab set, assessment vocabulary or claim discipline differs; the provenance strip and disabled live writes are the only additions. Replay is not a tab: it adds a controller and every tab renders the recorded Site history as of t – evidence available by t, configuration valid at t, managed-object state as of t, no future leakage, no recomputation with newer algorithms.

Primary interaction. Scrub or play time within the selected window; open any tab at that time; Open in Simulator Lab ↗ leaves the Site root for the Simulator Lab workspace (separate shell). Changing the time window in the top bar changes every time-dependent tab consistently.

8.17 Simulation truth vs AssetOps evidence (simulator / admin only)

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Sites ▾ Demo Maji-02 ▾ Performance ▾ Dispatch ▾ Compare with simulation truth: ON

Demo Maji-02 Mini-grid

Truth comparison · admin mode · SimulationTruth is never consumed by product analytics

site_id maji-mini-grid-001 SIMULATED Scenario MG-HYBRID-04 v4 · Contributing run(s) in window sim-2041 · As-of Aug 11 16:50

Open in Simulator Lab ↗

Live field v disabled

Overview

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Fuel

Renewable

Backup

ADMIN VIEW - hidden from operator roles

Product view: Candidate avoidable · conf. High · coverage 94%

Interval 14:20-16:50 - simulation truth vs AssetOps evidence

QUANTITY	SIMULATION TRUTH	ASSETOPS EVIDENCE	PRODUCT CLAIM	VERDICT
Cause of genset run	Forced policy (scenario event)	Inferred: support need ended	Candidate avoidable	Correct
Battery discharge capability	28 kW	22 kW observed; limit missing 13 min	Coverage 94%	Ceiling
Manual override	false	No record (observed absence)	Listed as alternative	Correct
Fuel movement (Aug 9)	EXTERNAL_ADJUSTMENT -80 L (truth)	Tank readings, genset consumption, delive...	Material unexplained variance, not theft	Correct
Tank sensor bias (from Aug 15)	+4%	Not yet detectable	—	Expecte

Used for false-positive checks, claim-ceiling validation and demonstrating why AssetOps refuses stronger claims. Truth values never enter Evidence, Findings or Financials

Figure 17. Simulation truth vs AssetOps evidence (simulator / admin only).

Content and evidence behaviour. SimulationTruth belongs to the SimulationRun / private simulator evaluation context; observable source evidence belongs to Site history. The comparison is a read-only overlay for simulator/admin roles; product analytics, evidence objects, operator UI and exports have no access to it.

Primary interaction. Toggle off returns to the normal product view; the Expectations layer of the scenario reads this comparison to pass or fail regression assertions. The truth values are the same ones visible in Simulator Lab > Devices & Sensors (8.13); they reach AssetOps only through this admin overlay.

8.18 Mini-grid site overview

Navigation state	Purpose
Sites → Maji Village → Overview	Summarize whether the mini-grid is fulfilling its operating objective and what needs attention now.

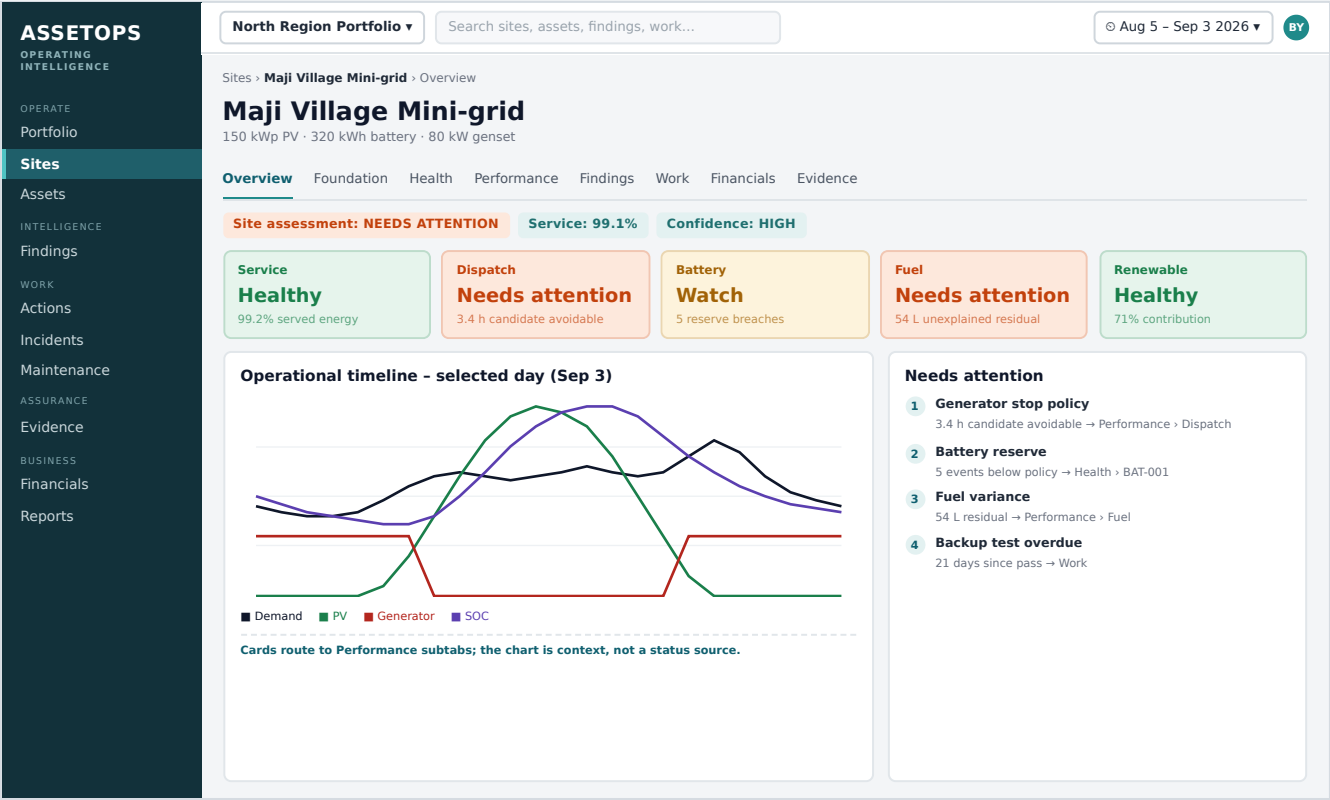


Figure 18. Mini-grid site overview.

Content and evidence behaviour. Cards use canonical assessments (Healthy, Watch, Needs attention, Degraded, Unknown) plus a concise reason; they do not mix opportunity/strength terms as status values. Assessment cards are coloured; metric cards elsewhere are neutral.

Primary interaction. Cards route to Performance, Health, Findings, Work, Financials or Evidence within the site.

8.19 Mini-grid site health

Navigation state	Purpose
Sites → Maji Village → Health	Provide a site-level asset-condition summary without duplicating asset-detail screens.

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Sites ▸ Maji Village Mini-grid ▸ Health

Maji Village Mini-grid

Health · site asset-condition summary

OverviewFoundationHealthPerformanceFindingsWorkFinancialsEvidence

Site asset assessment: WATCH4 monitored asset groupsSummary only - asset detail lives in Assets

Physical asset condition

ASSET / GROUP	ASSESSMENT	EVIDENCE	PRIMARY REASON	OPEN FINDINGS	OPEN WORK	
BAT-001 Battery	Watch	Medium-high	5 reserve breaches	2	0	Open asset →
GEN-001 Genset	Healthy	High	No condition issue	1	1	Open asset →
INV-001 PV inverter	Healthy	High	Available / no recurring alarm	0	0	Open asset →

Clicking an asset changes the root entity: left nav moves to Assets and the breadcrumb keeps the site.

Instrumentation / source health (data quality - not asset condition)

SOURCE	STATUS	QUALITY	REASON	OPEN WORK	
GW-001 Gateway	Online	Good	Clock sync +180 ms; buffer empty	0	Open asset →
Tank sensor	Stale 11 min	Degraded	Affects fuel claim ceiling	1	Site Evidence →

Source health uses Online / Stale / Offline and a quality grade, never the asset assessment vocabulary. Full source detail lives in Site Evidence.

Figure 19. Mini-grid site health.

Content and evidence behaviour. Two visually separate blocks: physical asset condition (canonical assessment vocabulary) and instrumentation/source health (Online / Stale / Offline + quality grade). Gateway 'Online' is never presented as equivalent to Battery 'Healthy'.

Primary interaction. Open asset links to the corresponding Asset Detail (root changes to Assets).

8.20 Mini-grid performance - Service

Navigation state	Purpose
Sites → Maji Village → Performance → Service	Answer whether required load service was delivered.

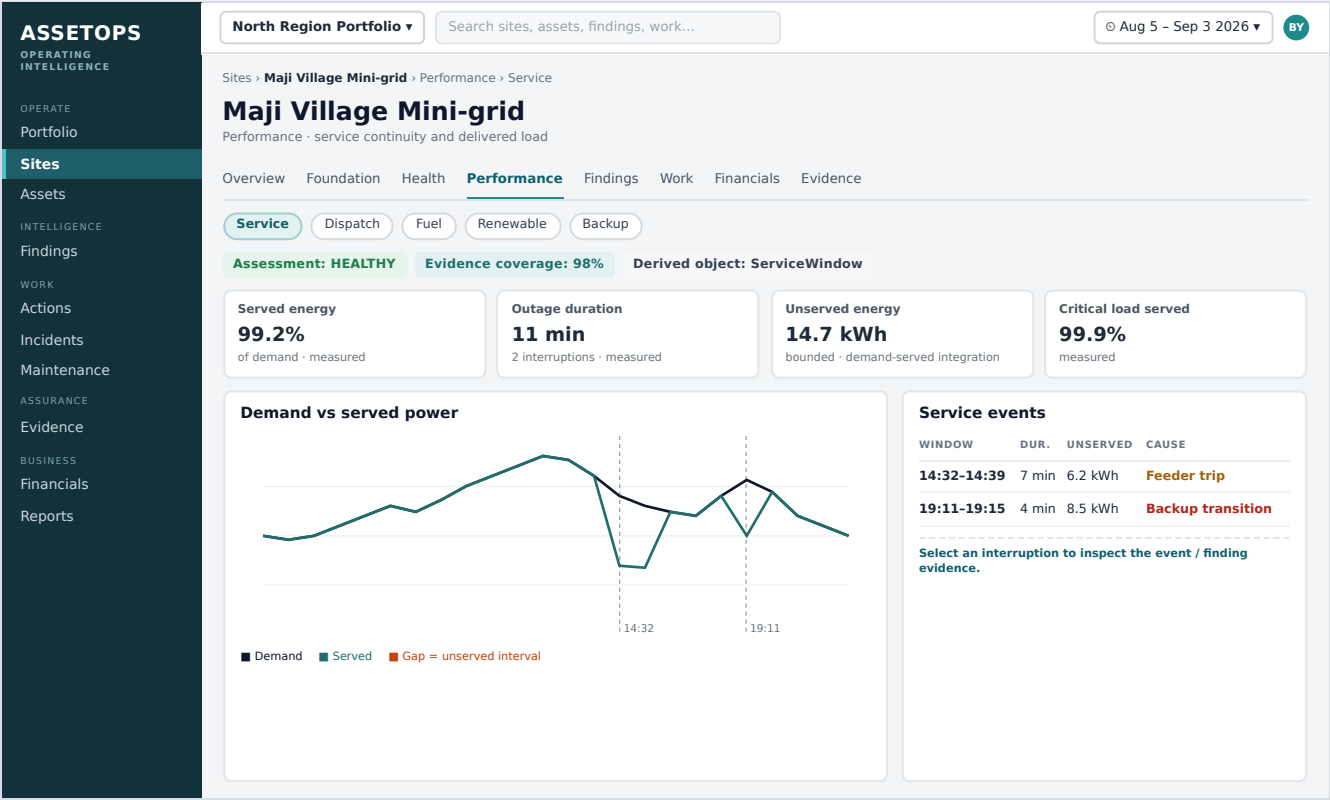


Figure 20. Mini-grid performance - Service.

Content and evidence behaviour. Based on ServiceWindow: served energy, outage duration, unserved energy and coverage. Insufficient evidence cannot be shown as healthy; coverage below threshold degrades the assessment to Unknown.

Primary interaction. Select an interruption to inspect the event/finding evidence.

8.21 Mini-grid performance - Dispatch

Navigation state	Purpose
Sites → Maji Village → Performance → Dispatch	Explain why sources operated and whether generator intervals were feasible to avoid.

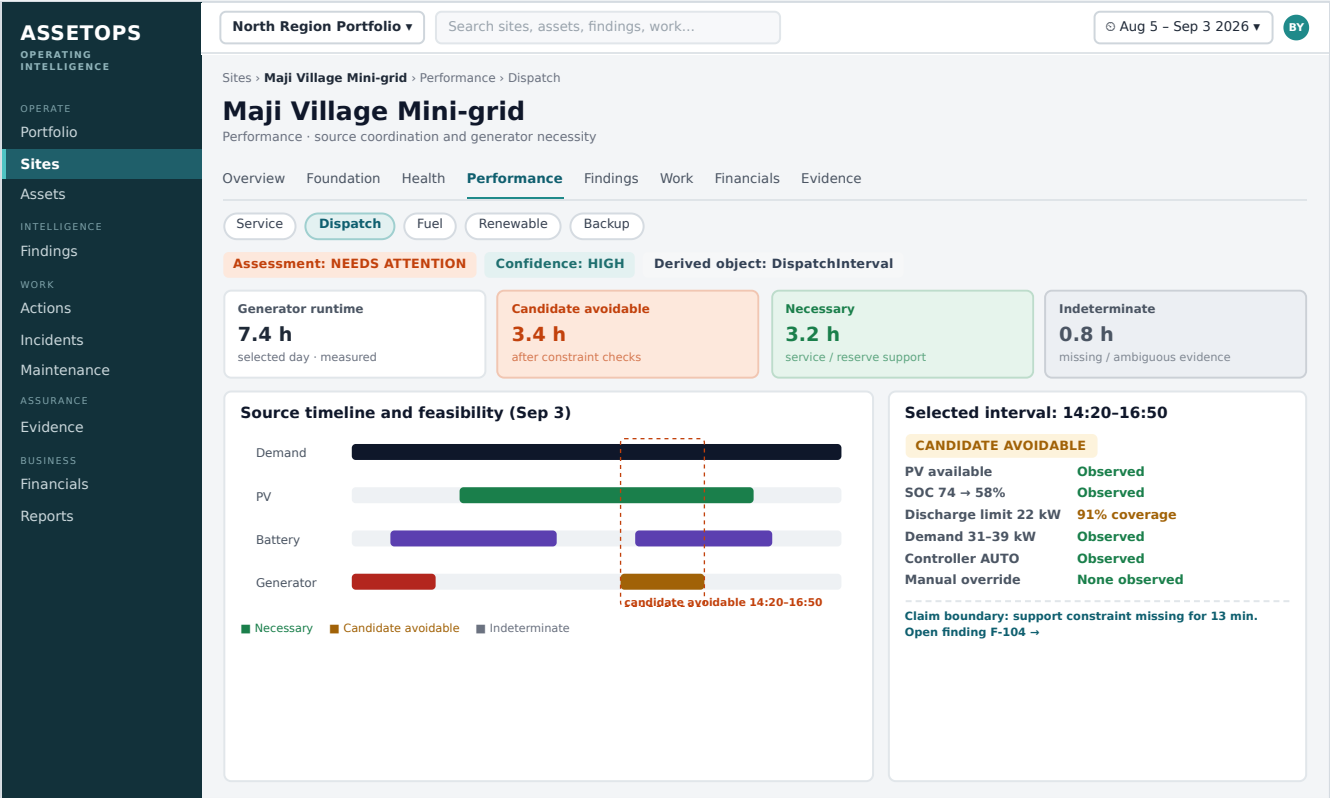


Figure 21. Mini-grid performance - Dispatch.

Content and evidence behaviour. Dispatch intervals are Necessary, Candidate Avoidable or Indeterminate after PV, battery, demand, controller and policy constraints are checked. Indeterminate is shown in neutral grey so it cannot be read as either good or bad.

Primary interaction. Select an interval to open its finding/evidence.

8.22 Mini-grid performance - Fuel

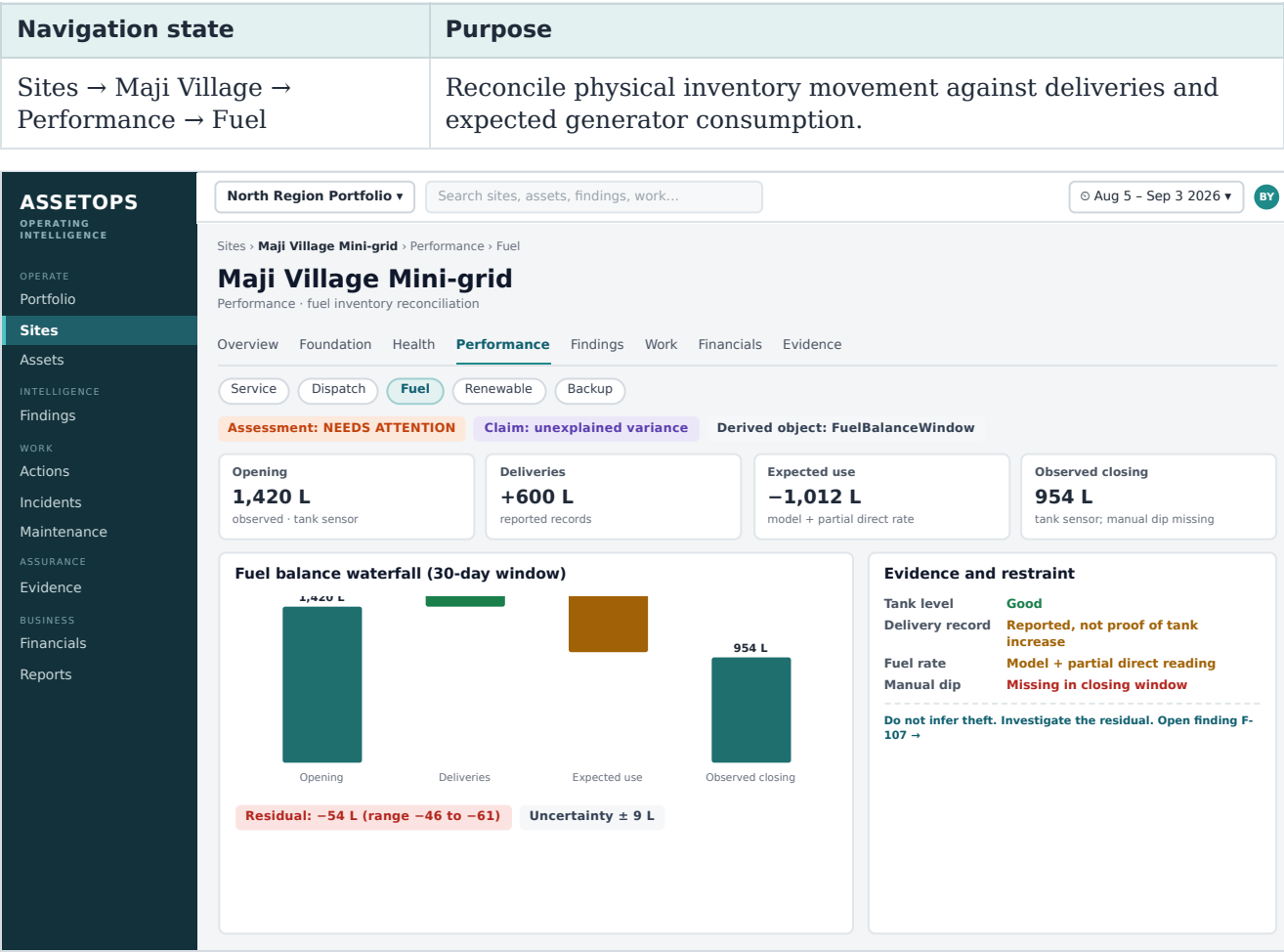


Figure 22. Mini-grid performance - Fuel.

Content and evidence behaviour. Displays opening + deliveries – expected use – closing with uncertainty. The result is an unexplained residual/variance, never a theft conclusion by itself. Metric cards are neutral; only the residual chip carries a status colour.

Primary interaction. Open the residual finding or inspect delivery/dip evidence.

8.23 Mini-grid performance - Renewable

Navigation state	Purpose
Sites → Maji Village → Performance → Renewable	Identify renewable energy that could feasibly have displaced higher-cost supply.

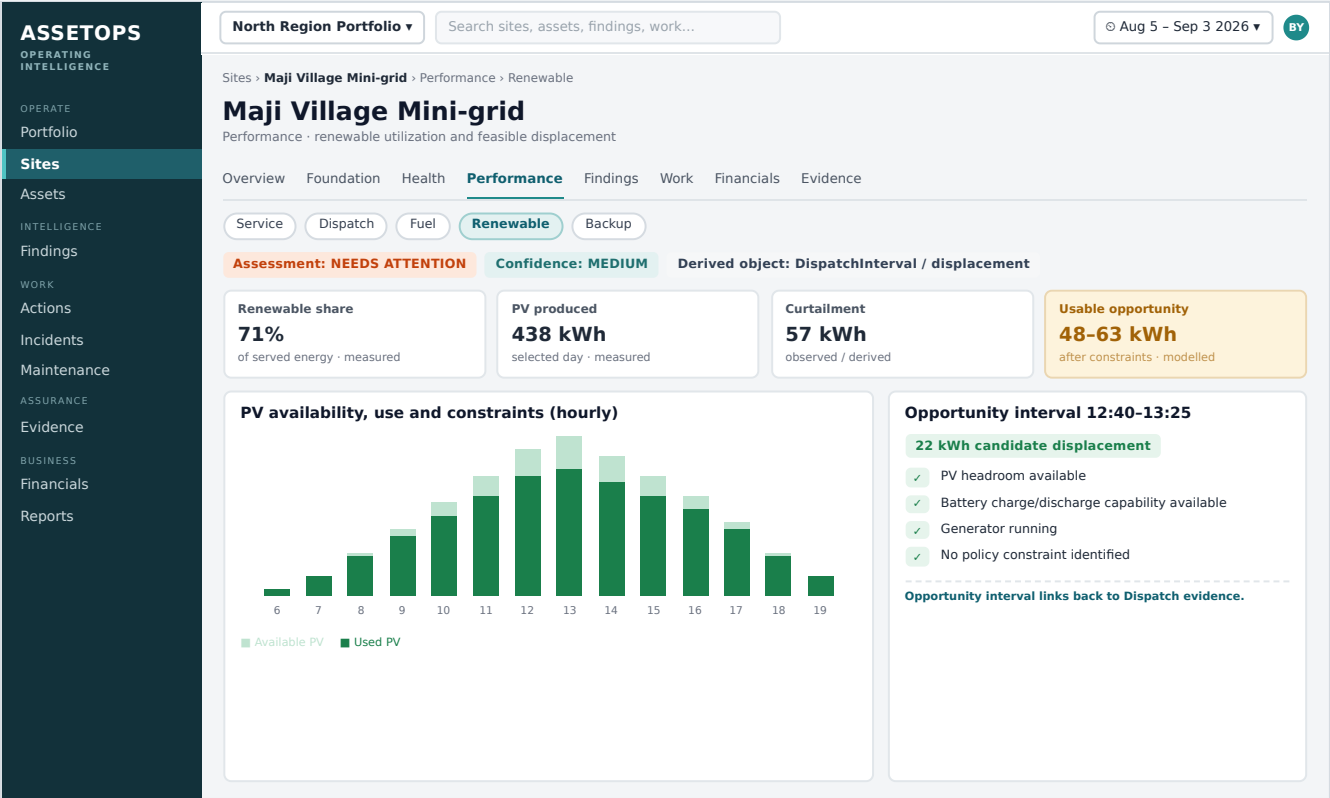


Figure 23. Mini-grid performance – Renewable.

Content and evidence behaviour. Opportunity is shown only after demand, battery limits, generator/grid state, PV availability and policy constraints permit it.

Primary interaction. Opportunity interval links back to Dispatch evidence.

8.24 Mini-grid performance - Backup

Navigation state	Purpose
Sites → Maji Village → Performance → Backup	Measure whether backup restored required electrical service during outages.

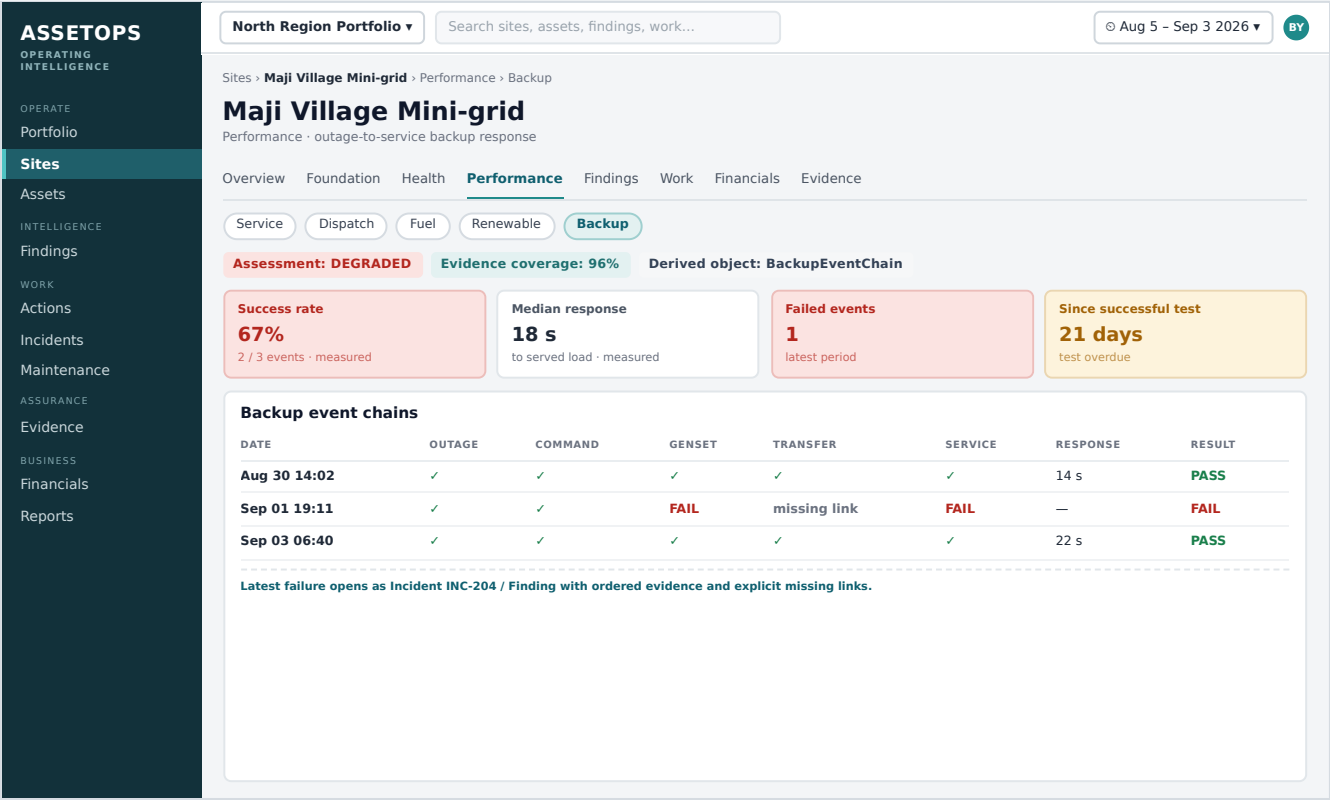


Figure 24. Mini-grid performance - Backup.

Content and evidence behaviour. Uses BackupEventChain: outage, command, genset/transfer, served load, response time and explicit missing links.

Primary interaction. Failed events open as findings/incidents (root changes to Incidents).

8.25 Site findings (scoped index)

Navigation state	Purpose
Sites → Maji Village → Findings	Show the findings that concern this site without creating a second findings list or status model.

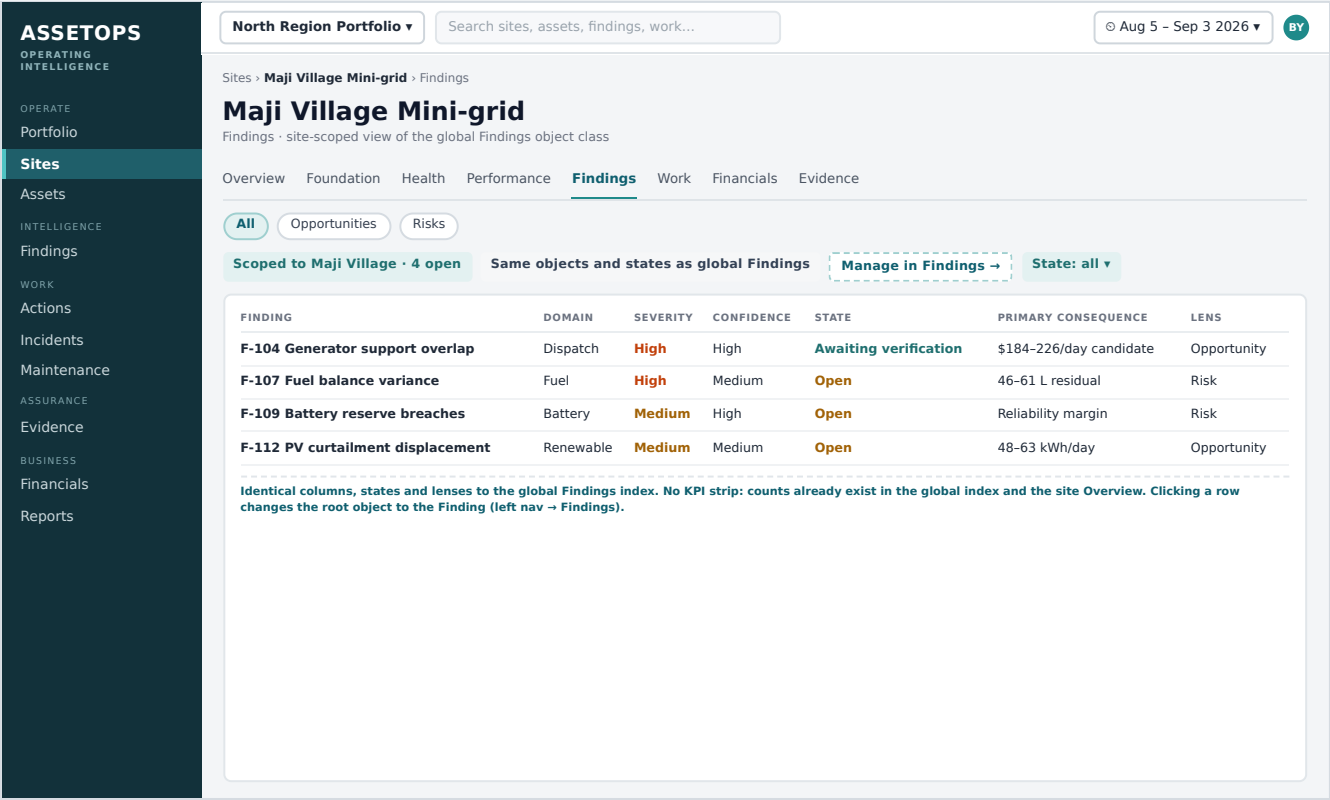


Figure 25. Site findings (scoped index).

Content and evidence behaviour. This tab is a read-only, site-filtered index over the global Findings object class. It carries no KPI strip, no separate assessment and no second status model; the site's analytical content lives in Performance and Health. The only site-specific column is Domain, which links back to the Performance subtab that produced the finding.

Primary interaction. Clicking a finding opens Finding Detail with Findings highlighted; 'Manage in Findings' opens the global index pre-filtered to the site.

8.26 Site work (scoped index: Actions | Incidents | Maintenance)

Navigation state	Purpose
Sites → Maji Village → Work → Actions	Show the managed work attached to this site, mirrored from the three Work object classes.

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BY

Sites ▸ Maji Village Mini-grid ▸ Work ▸ Actions

Maji Village Mini-grid

Work ▸ site-scoped view of Actions, Incidents and Maintenance

OverviewFoundationHealthPerformanceFindingsWorkFinancialsEvidence

Actions (3)Incidents (1)Maintenance (2)

Scoped to Maji Village · 3 open · 1 in verification window

Same objects and states as global Actions

Manage in Actions →

ACTION	SOURCE	OWNER	DUE	STATE	SUCCESS CRITERION
ACT-0048 Reduce avoidable generator r...	Finding F-104	Controls lead	Sep 10	Verification window	≥ 25% runtime reduction, no trade-off
ACT-0053 Complete closing fuel dip	Finding F-107	Operator	Sep 6	Open	Independent closing inventory evidence
ACT-0057 Run witnessed backup test	Foundation readiness	Site lead	Sep 8	Open	Automatic start + service within policy

The subtab row mirrors the WORK group of the left navigation one-to-one (Actions | Incidents | Maintenance), so a user always knows where the same record is managed globally.

Figure 26. Site work (scoped index: Actions | Incidents | Maintenance).

Content and evidence behaviour. The site Work tab is the scoped counterpart of the left-nav WORK group. Its one subtab row uses the same three names as the global menu, and each subtab is a filtered index with identical columns and lifecycle states. Counts in the subtab labels replace a KPI strip.

Primary interaction. Clicking a row changes the root object (Actions, Incidents or Maintenance highlighted); creating work from here uses the global create flow with the site pre-filled.

8.27 Battery asset - Health

Navigation state	Purpose
Assets → BAT-001 → Health	Assess observable battery condition without unsupported remaining-life prediction.

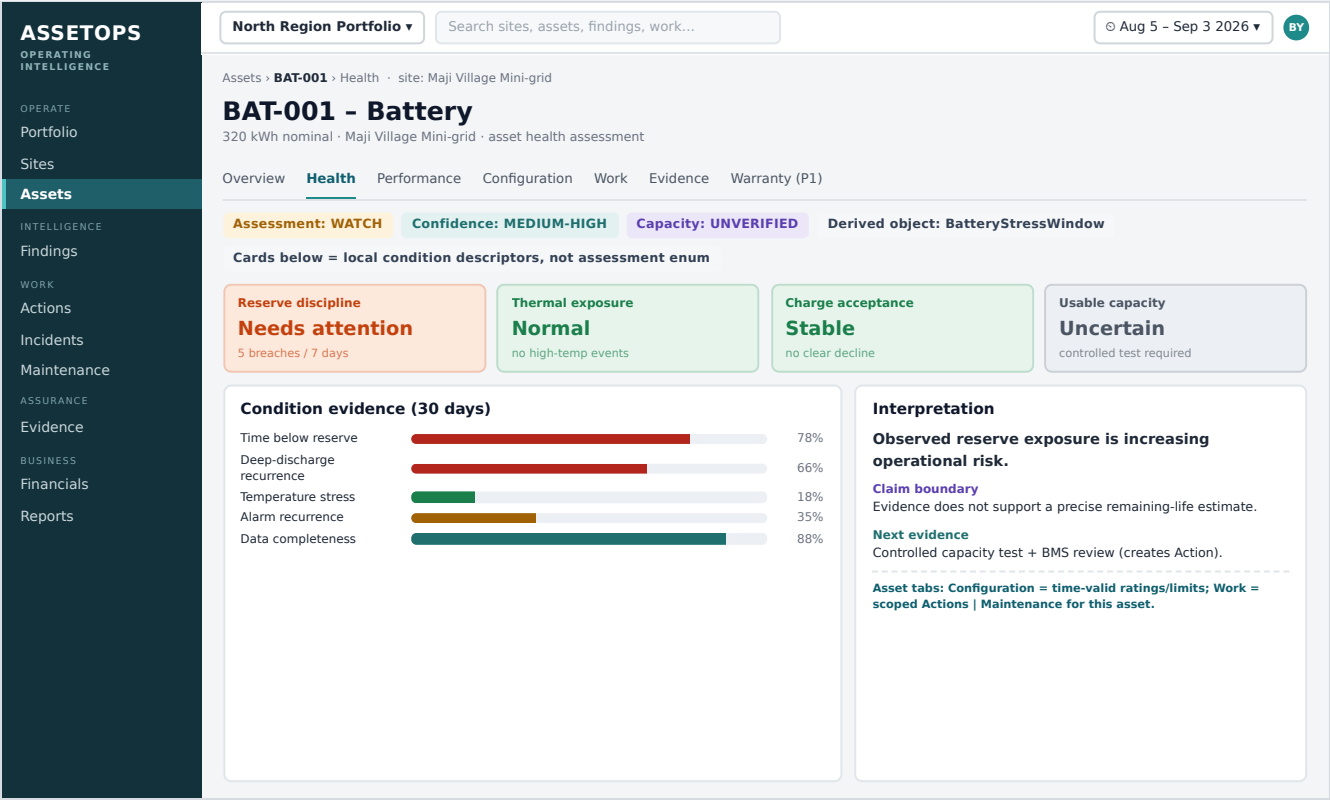


Figure 27. Battery asset – Health.

Content and evidence behaviour. Uses reserve breaches, power, temperature/alarms and capacity evidence. Capacity is explicitly unverified until controlled evidence exists. The asset's 'Maintenance' tab from v5 is renamed 'Work' so scoped-index tabs always carry the global object-class name.

Primary interaction. Navigate to Performance, Configuration, Work, Evidence or Warranty within the same asset; the site link in the breadcrumb returns to Sites.

8.28 Cold-room site overview

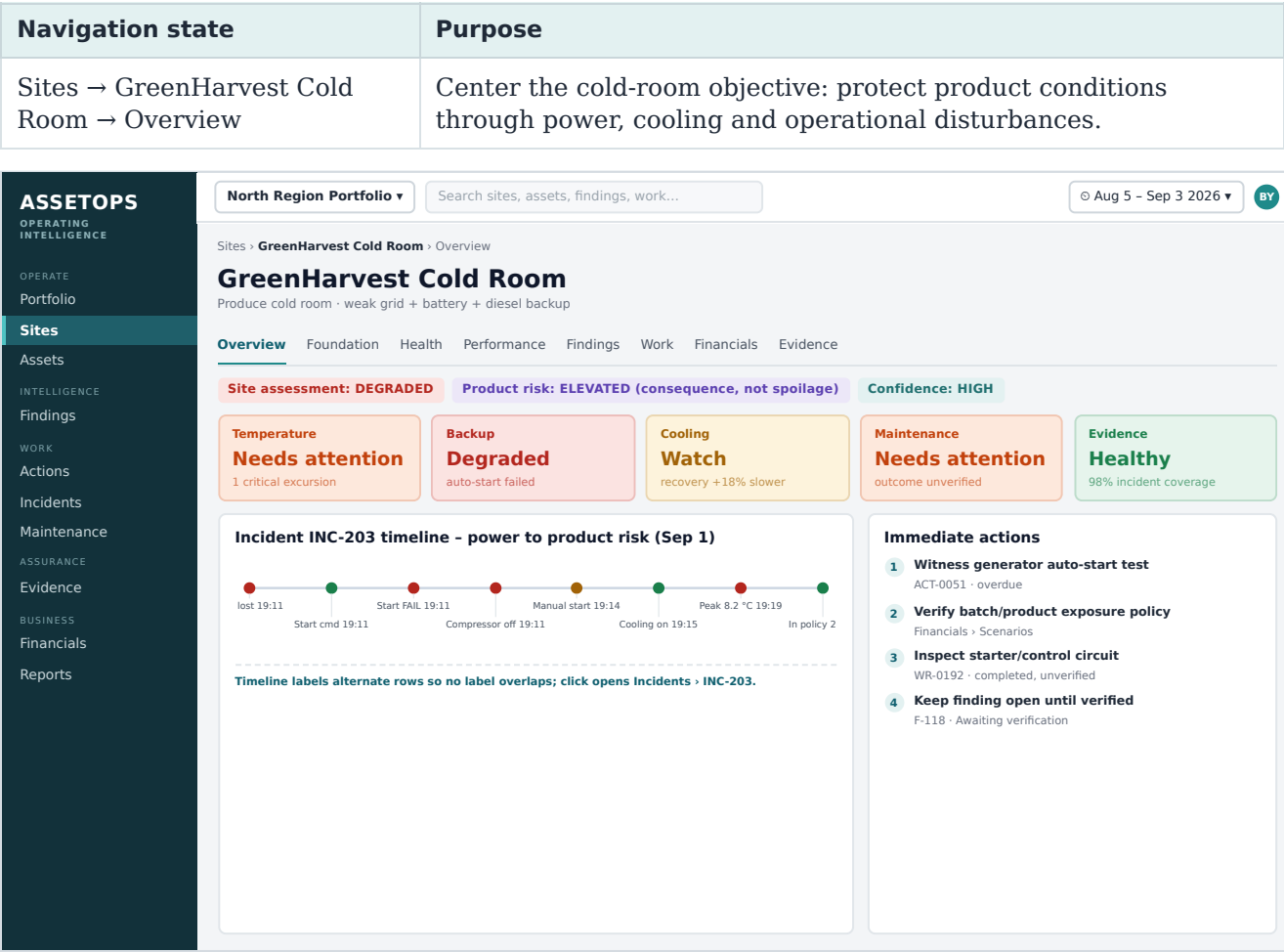


Figure 28. Cold-room site overview.

Content and evidence behaviour. Separates Temperature, Backup, Cooling, Maintenance and Evidence assessments; product risk is consequence context, not a spoilage determination, so it is a purple consequence chip rather than an assessment card.

Primary interaction. Open the relevant Performance view, finding, incident or work item.

8.29 Cold-room performance - Temperature

Navigation state	Purpose
Sites → GreenHarvest → Performance → Temperature	Determine whether chamber conditions stayed within configured product policy and characterize excursions.

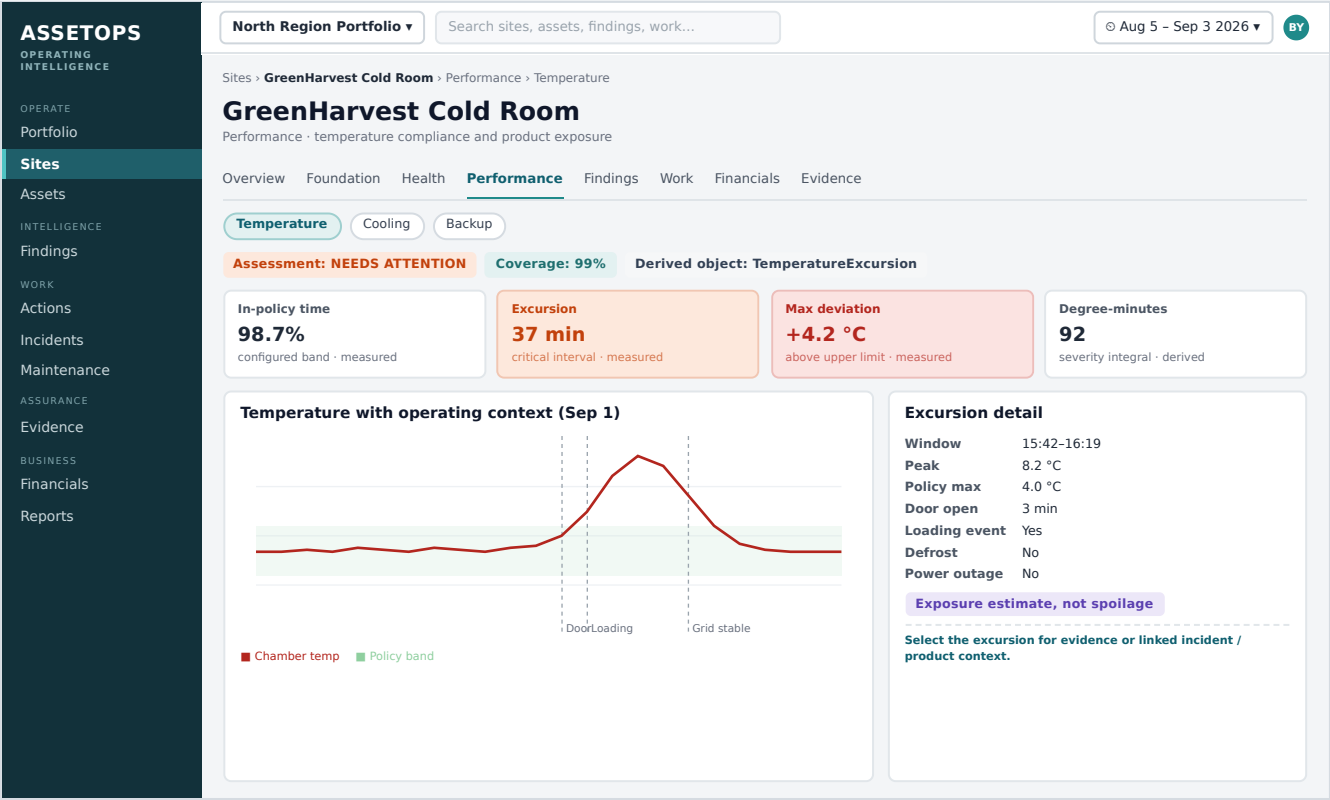


Figure 29. Cold-room performance - Temperature.

Content and evidence behaviour. Shows time in policy, duration, max deviation and degree-minutes with door/loading/power context. Product exposure remains bounded and conditional.

Primary interaction. Select the excursion for evidence or linked incident/product context.

8.30 Cold-room performance - Cooling

Navigation state	Purpose
Sites → GreenHarvest → Performance → Cooling	Assess whether cooling recovery is degrading beyond explainable operating context.

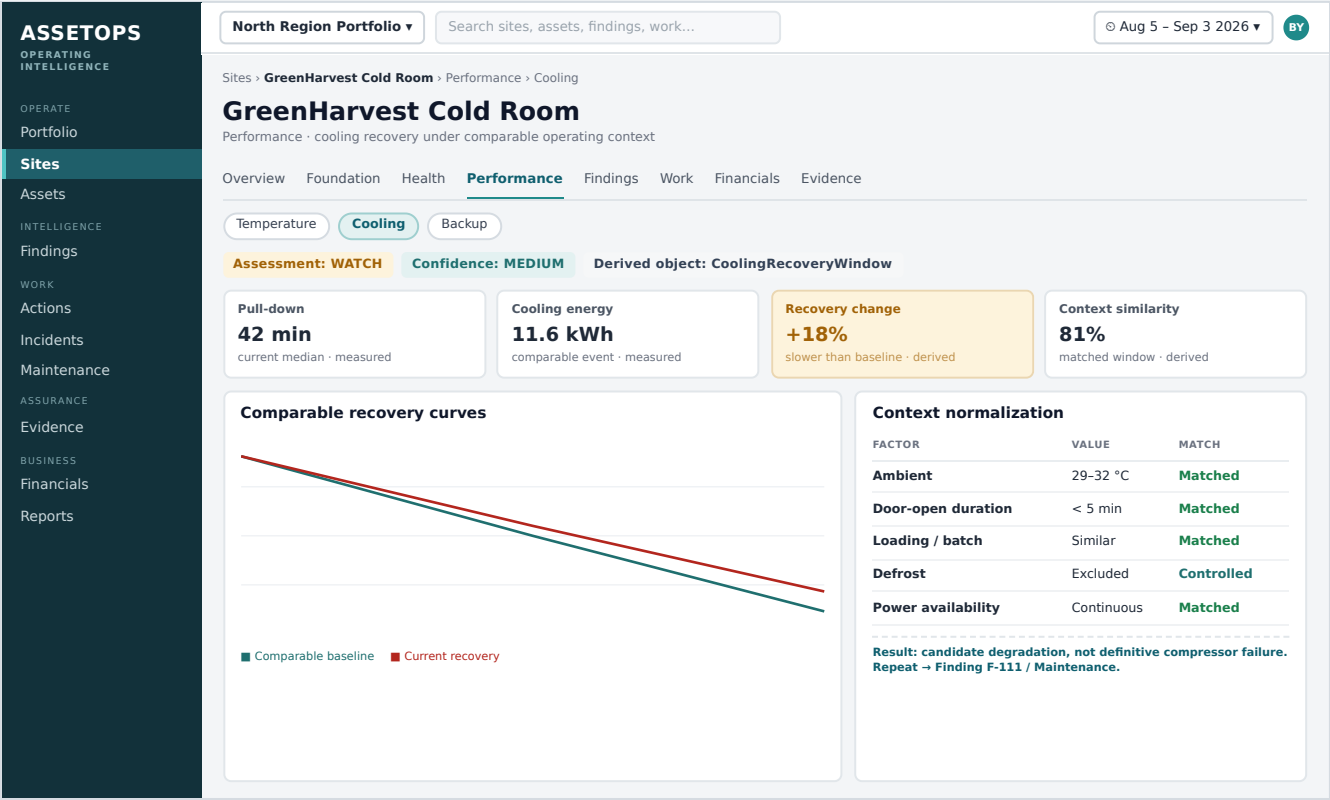


Figure 30. Cold-room performance - Cooling.

Content and evidence behaviour. Comparable-window analysis controls ambient, door/open time, loading/batch, defrost and energy availability. Output is a degradation candidate, not automatic compressor failure.

Primary interaction. Open the finding or maintenance action if repeated.

8.31 Cold-room performance - Backup

Navigation state	Purpose
Sites → GreenHarvest → Performance → Backup	Reconstruct grid-outage to cooling-service recovery and distinguish backup failure from refrigeration failure.

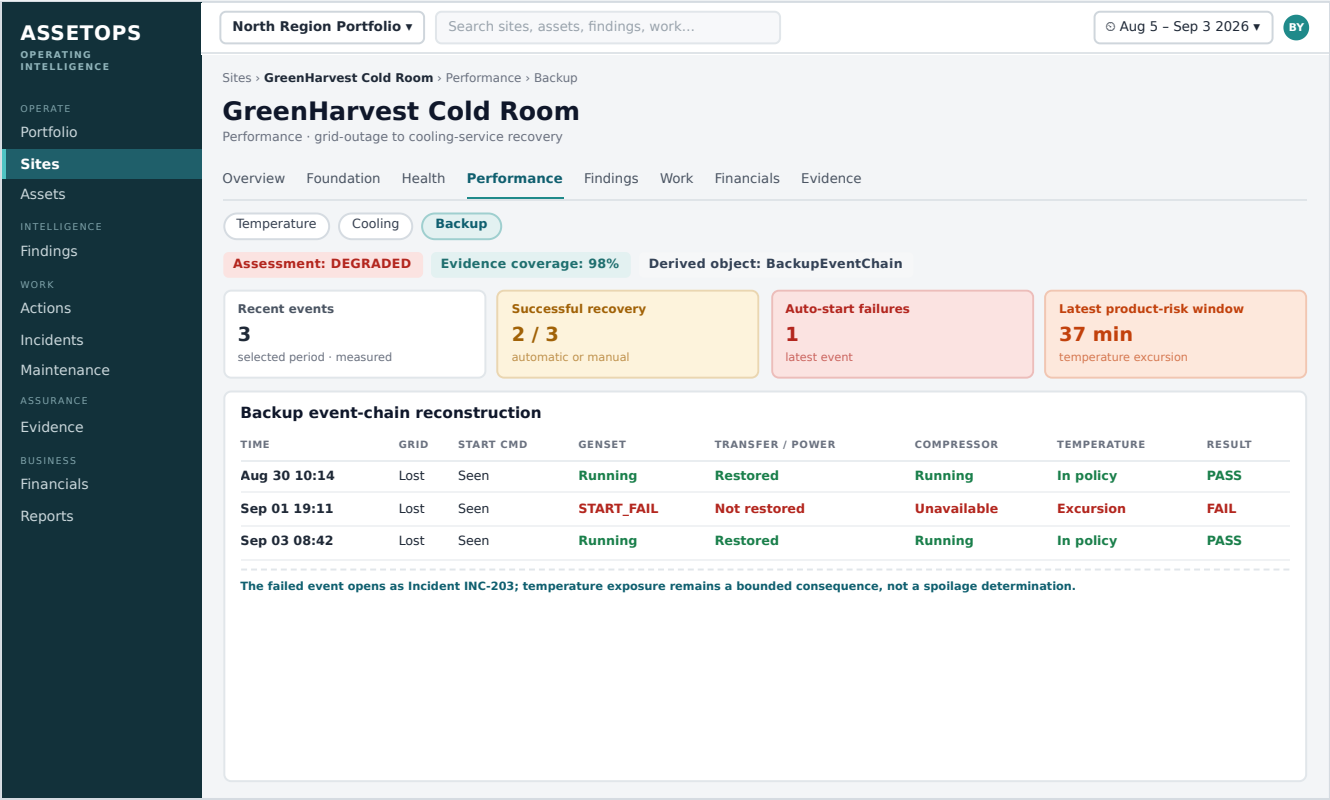


Figure 31. Cold-room performance - Backup.

Content and evidence behaviour. Links grid state, start command, genset, transfer/power path, compressor availability and temperature outcome. Temperature excursion is a consequence in the chain.

Primary interaction. Failed event opens Incident Detail and linked finding.

8.32 Finding detail - Overview

Navigation state	Purpose
Findings → F-104 → Overview	Present one defensible analytical conclusion.

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BY

Findings · F-104 · Overview · site: Maji Village Mini-grid

F-104 Generator remained online after support need ended

Maji Village Mini-grid · Dispatch · Sep 3 14:20-16:50

Overview

Evidence

Alternatives

Recommendation

History

Severity: HIGH

Confidence: HIGH

State: AWAITING VERIFICATION

Claim: CANDIDATE AVOIDABLE

Lens: Opportunity

Candidate avoidable runtime

3.4 h

interval · derived

Fuel consequence

34-42 L

modelled range

Daily value

\$184-226

modelled · not realized

Evidence coverage

94%

13 min discharge-limit gap

Bounded claim

Generator remained online while PV and battery capability appeared sufficient to support observed demand.

WHY THIS MATTERS
Recurring pattern on 9 of last 14 days; potential fuel consequence 34-42 L per interval.

CLAIM BOUNDARY
Support constraint evidence missing for 13 min; claim stays Candidate until PCS discharge limit is exposed.

Where to look next

1 Evidence tab

Full evidence chain + claim evidence drawer

2 Alternatives tab

4 alternative explanations, ranked; what would discriminate

3 Recommendation tab

Review stop policy; success = lower runtime, no reserve/service trade-off

4 Linked action

ACT-0048 · verification window to Sep 10

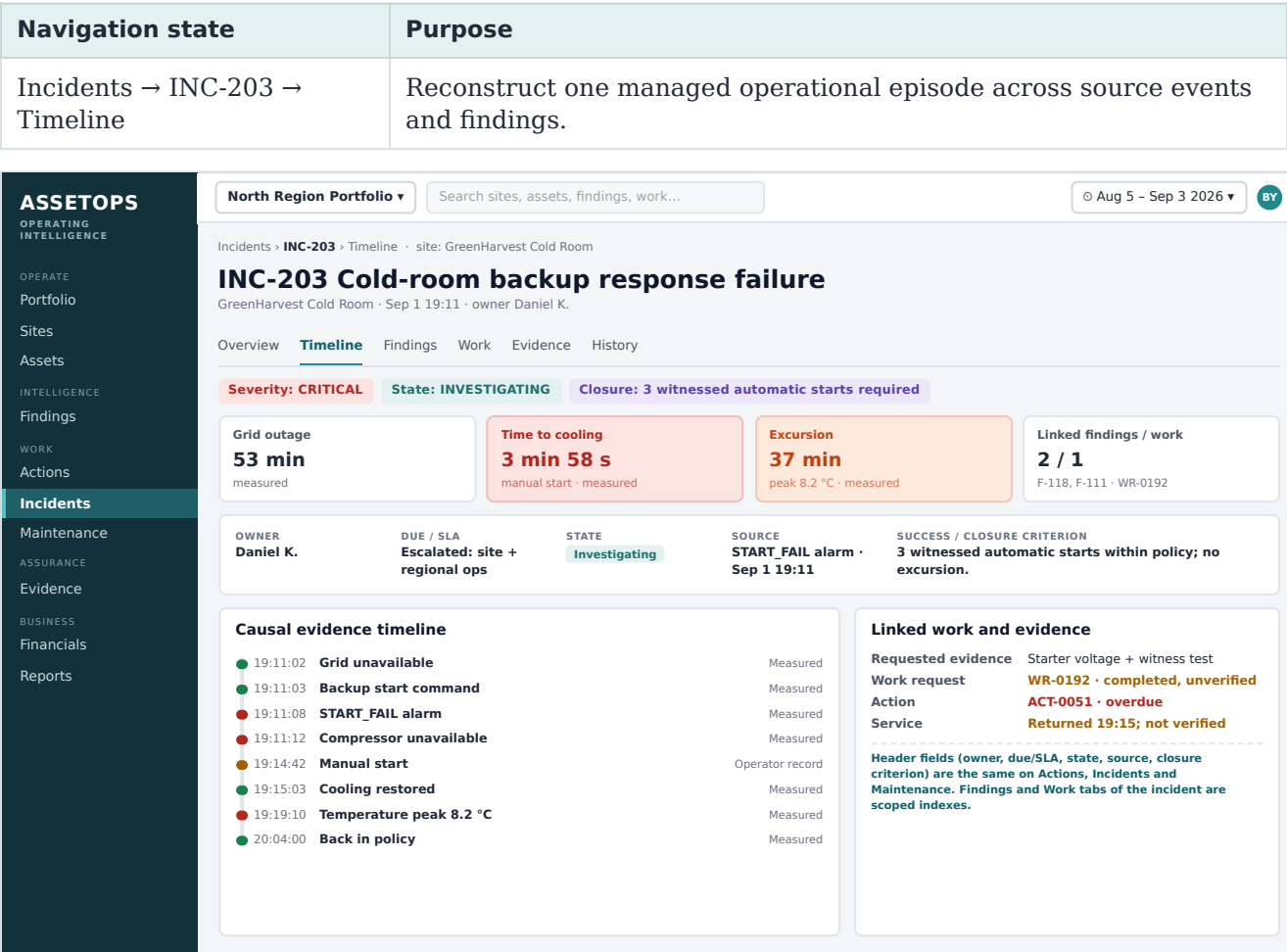
Overview no longer duplicates the Evidence, Alternatives and Recommendation tabs; it states the claim, consequence and boundary.

Figure 32. Finding detail – Overview.

Content and evidence behaviour. Overview separates observed facts, inference, consequence, confidence and claim boundary. In v5 the Overview repeated the evidence chain, alternatives and recommendation that have their own tabs; v6 keeps Overview to the bounded claim and routes to the tabs.

Primary interaction. Evidence opens claim-specific provenance; an accepted recommendation creates an Action (root changes to Actions).

8.33 Incident / case detail - Timeline



8.34 Maintenance work detail - Verification

Navigation state

Maintenance → WR-0192 → Verification

Purpose

Determine whether completed maintenance changed the target operating behaviour.

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BY

Maintenance > WR-0192 > Verification > incident: INC-203

WR-0192 Replace starter relay

GreenHarvest Cold Room · CMP-003 / genset starter · corrective

Overview

Plan

Execution

Evidence

Verification

History

Work status: COMPLETED

Outcome: UNVERIFIED

Finding F-118 remains OPEN

OWNER

Tech A

DUE / SLA

Witness test by Sep 8

STATE

Completed - unverified

SOURCE

Incident INC-203 / Finding F-118

SUCCESS / CLOSURE CRITERION

Automatic start restores cooling within policy on 3 consecutive witnessed outages; no START_FAIL recurrence.

Lifecycle

Finding

Triage

Assigned

Visit

Completed

Verified

Work evidence (activity)

ITEM	STATUS
Technician notes	Captured
Starter relay replaced	Captured
Part serial / custody	Partial
Automatic start witness	Missing
Return-to-service reading	Captured

Measured impact (outcome)

METRIC	RESULT	NOTE
Backup start	Not yet tested	Witness test pending
Cooling recovery	Degraded	Not comparable yet
Energy intensity	Elevated	Monitoring
Recurrence	Unknown	7-day window open

Closed work order ≠ resolved finding. Verification passes only when declared evidence and success criterion are satisfied.

Figure 34. Maintenance work detail – Verification.

Content and evidence behaviour. Work evidence (activity) and post-action performance (outcome) are separate panels on one Verification tab. v5 had both an Impact tab and a Verification tab on work requests while Actions had only Verification; v6 gives Actions and Maintenance work requests the same six tabs.

Primary interaction. Verification remains pending until declared evidence and success criteria are satisfied; outcome can be Verified, Ineffective, Inconclusive or Reopened.

8.35 Site evidence and claim readiness

Navigation state	Purpose
Sites → Maji Village → Evidence	State what AssetOps can substantiate at this site and what additional evidence would improve claims.

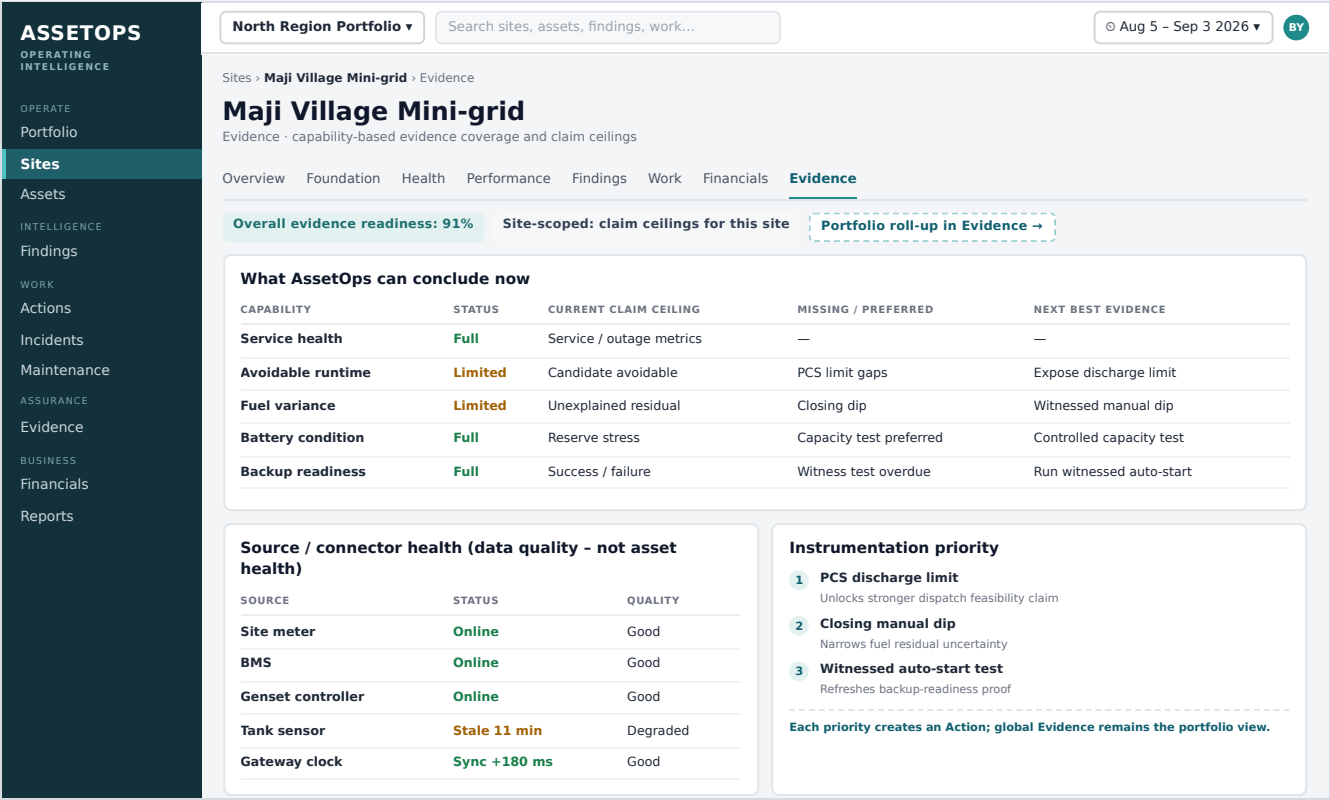


Figure 35. Site evidence and claim readiness.

Content and evidence behaviour. Each capability shows Full/Limited/None/N-A, current claim ceiling, missing/preferred evidence and next-best evidence. Gateway/source health is separate from physical asset health. This is site-owned content (claim ceilings), not a duplicate of the global Evidence roll-up.

Primary interaction. Drill into a capability or specific source; the global Evidence link opens the portfolio roll-up (left nav moves to Evidence).

8.36 Site financials

Navigation state

Sites → Maji Village → Financials → Overview

Purpose

Translate this site's technical findings into bounded economic consequence.

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BY

Sites ▾ Maji Village Mini-grid ▾ Financials ▾ Overview

Maji Village Mini-grid

Financials - technical behaviour translated to business consequence

Overview

Foundation

Health

Performance

Findings

Work

Financials

Evidence

Overview

Drivers

Assumptions

Scenarios

Financial consequence identified: 4 lines

Data confidence: MEDIUM

No financial assessment - status lives on the source findings

Portfolio roll-up in Financials →

Avoidable OPEX

\$3.8-4.7k/mo

modelled

Revenue / service

\$0.76-0.92k/mo

measured × tariff

Asset exposure

Scenario

not a life prediction

Assurance cost

\$0.22-0.27k/mo

modelled

Value-driver bridge

TECHNICAL BEHAVIOUR	FINANCIAL MECHANISM	BUSINESS CONSEQUENCE	BASIS	CONF.	SOURCE
Candidate avoidable genset runtime	Fuel + maintenance hours	Higher recurring OPEX	Modelled range	Med-high	F-104
Battery reserve violations	Reduced reliability margin	Replacement exposure / outage risk	Scenario	Medium	F-109
Unserved mill demand	Energy not sold	Lost revenue	Tariff × measured kWh	High	Service
Fuel residual	Unexplained inventory movement	Cash / accountability exposure	Measured + modelled	Medium	F-107

Discipline: mark measured / modelled / extrapolated / scenario-based; avoid double counting; show trade-offs. Clicking a value returns to its source finding.

Figure 36. Site financials.

Content and evidence behaviour. Financial categories trace to technical mechanisms and business facts. Basis and confidence stay visible, and overlapping consequences are not summed. Financial pages carry no assessment of their own: 'Needs attention' has no deterministic financial definition, so v6.1 replaces it with a count of consequence lines and leaves status on the source findings. Drivers / Assumptions / Scenarios are local financial subtabs.

Primary interaction. Clicking a value returns to its source finding; the global Financials link opens the portfolio roll-up.

8.37 Claim evidence drawer

Navigation state	Purpose
Findings → F-104 → Evidence → drawer	Inspect exact provenance of a material claim without leaving the current analytical context.

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BY

Findings · F-104 · Evidence

F-104 Generator remained online after support need ended

Evidence · full evidence chain; drawer opened from the claim

Overview

Evidence

Alternatives

Recommendation

History

Claim: 3.4 h candidate avoidable runtime

Confidence: HIGH

Coverage: 94%

Evidence chain

INPUT	OBSERVATION	BASIS	
PV capacity	Available power exceeded load margin	Measured	Inspect →
Battery SOC	Above reserve threshold	Measured	Inspect →
Battery limits	Mostly available; 13 min missing	Measured (gap)	Inspect →
Generator	Running at low loading	Measured	Inspect →
Controller mode	AUTO	Measured	Inspect →
Context	No manual override recorded	Observed absence	Inspect →

Basis vocabulary: Measured / Modelled / Extrapolated / Scenario / Observed absence / Unknown.

Evidence drawer

Candidate avoidable runtime · 14:20-16:50

DispatchInterval

ALGORITHM

dispatch-feasibility-v3

INPUTS

SIGNAL	COV.	QUALITY
MG-004 PV active power	100%	Good
MG-008 Battery SOC	100%	Good
MG-011 Discharge limit	91%	13 min missing
MG-014 Generator state	100%	Good
MG-025 Controller mode	100%	Good

CONFIGURATION AT TIME

Battery reserve 25% · config-v18

Genset min runtime 10 min · config-v11

PROVENANCE

Mapping bms-map-v3 · Gateway gw-config-2026-09

Run sim-20260903-041

APPROVAL / OVERRIDE HISTORY

None recorded

Show raw evidence →

Figure 37. Claim evidence drawer.

Content and evidence behaviour. Shows derived object, time window, raw/normalized evidence refs, coverage/quality, configuration-at-time, mapping/algorithm versions and approval/override history. The drawer is claim-specific lineage; the Evidence page is capability readiness.

Primary interaction. Raw evidence inspection is available from the drawer; the drawer never changes the root object.

8.38 Action - Verification

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BY

Actions > ACT-0048 > Verification · finding: F-104

ACT-0048 Reduce avoidable generator runtime

Maji Village Mini-grid · source F-104 · owner site controls lead

Overview

Plan

Execution

Evidence

Verification

History

State: VERIFICATION WINDOW · day 5 of 7

Outcome: PROVISIONAL PASS

Final outcome set on Sep 10 when the window closes

OWNER
Site controls lead

DUE / SLA
Sep 10 (window close)

STATE
Verification window

SOURCE
Finding F-104

SUCCESS / CLOSURE CRITERION
Reduce low-load generator runtime by ≥ 25% without increasing battery reserve breaches or unserved critical load.

Before / after (7-day pre-window vs post-window to date)

METRIC	BEFORE	AFTER (5 d)	ROLE	RESULT
Candidate avoidable runtime	3.4 h/day	1.9 h/day	Target	PASS so far (–44%)
Reserve breaches	5 / 7 days	0 / 5 days	Guardrail	PASS so far
Unserved critical load	0.3%	0.3%	Guardrail	PASS so far
Fuel use	122 L/day	108 L/day	Consequence	Consistent

Close-out evidence

Configuration change

Captured · CHG-0019

7-day post window

Day 5 of 7 - in progress

Gateway coverage

97%

No override confounder

Observed

Recurrence check

None to date

State becomes Verified only when the window closes with all criteria passing; that closes F-104. A failed or inconclusive result keeps or reopens it.

Figure 38. Action - Verification.

Content and evidence behaviour. Compares the target metric and guardrails over a post-action window, checks coverage/confounders and recurrence, and resolves only when criteria pass. State (Verification window) and outcome (Provisional pass) now agree: a final Verified / Ineffective / Inconclusive outcome is set only when the window closes. The same tab and panel layout is used by Maintenance work requests.

Primary interaction. Verified outcome may close the originating finding; failed/inconclusive results keep or reopen it.

9. Screen-to-evidence traceability summary

Screen / view	Route / root object	Primary derived / decision object	Claim / content discipline
Portfolio command center	Portfolio	Assessment + ranked finding queue	Deterministic roll-up; Unknown is not Healthy; opportunities are evidence-backed, not yet realized value
Sites / Assets / Findings / Actions / Incidents / Maintenance queues	Corresponding global collection	Index / filter over canonical objects	Indexes create no duplicate domain objects or alternate status models; Opportunities/Risks are lenses of Findings
Portfolio Evidence	Evidence	CapabilityEvidenceStatus roll-up	Portfolio scope only; drill to Site Evidence for site claim ceilings
Portfolio Financials	Financials	Non-overlapping consequence roll-up	Preserve measured/modelled/scenario basis; never double count
Site Foundation	Sites → Site → Foundation	Readiness/completeness rules	Analytics enabled only to evidence readiness; configuration is time-valid
Configuration change effect	Sites → Site → Foundation → Changes	Pre/post comparable-window outcome	Retain/retune/revert uses target + guardrail metrics
Site Health	Sites → Site → Health	Asset assessment summary	Summary only; asset click transfers to Assets
Site Findings	Sites → Site → Findings	Filtered Finding index	Scoped index; same states/lenses; no KPI strip; row click transfers to Findings
Site Work	Sites → Site → Work	Filtered Action / Incident / Work-request indexes	Subtab row mirrors WORK nav group; row click transfers to the object class
Service	Sites → Site → Performance → Service	ServiceWindow	Coverage constrains health; unserved energy from demand-served integration
Dispatch	Sites → Site → Performance → Dispatch	DispatchInterval	Necessary / Candidate avoidable / Indeterminate after feasibility and policy checks
Fuel	Sites → Site → Performance → Fuel	FuelBalanceWindow	Opening + deliveries – expected use – closing with uncertainty; unexplained variance, not theft
Renewable	Sites → Site → Performance → Renewable	DispatchInterval / displacement logic	Opportunity only when demand, storage, source and policy constraints permit

Mini-grid Backup	Sites → Site → Performance → Backup	BackupEventChain	Outage-to-served-load chain with response time and missing links
Battery Health	Assets → BAT-001 → Health	BatteryStressWindow + asset context	Observable condition/stress; no precise remaining-life claim
Temperature	Sites → Cold room → Performance → Temperature	TemperatureExcursion	Excursion and potential exposure; no automatic spoilage determination
Cooling	Sites → Cold room → Performance → Cooling	CoolingRecoveryWindow	Context-normalized degradation candidate, not automatic compressor failure
Cold-room Backup	Sites → Cold room → Performance → Backup	BackupEventChain	Grid → command → power path → compressor → temperature consequence
Finding Detail	Findings → Finding	Finding object	Overview = bounded claim + boundary; Evidence / Alternatives / Recommendation are tabs, not repeated
Incident Detail	Incidents → Incident	Incident / case timeline	Group related source events/findings without erasing provenance
Maintenance Work Verification	Maintenance → Work Request → Verification	VerificationOutcome / pre-post comparison	Completed work ≠ resolved finding; activity evidence and outcome are separate panels
Site Evidence	Sites → Site → Evidence	CapabilityEvidenceStatus	Full/Limited/None/N-A plus current claim ceiling and next-best evidence; links up to portfolio Evidence
Site Financials	Sites → Site → Financials	Consequence model	Technical bridge + business facts + basis/confidence; local subviews Drivers/Assumptions/Scenarios; links up to portfolio Financials
Evidence Drawer	Finding → Evidence drawer	Claim lineage	Exact refs, quality/coverage, configuration-at-time, mapping/algorithm version, approval history; inspection only
Action Verification	Actions → Action → Verification	VerificationOutcome	Resolve only when declared criterion and guardrails pass; otherwise inconclusive/reopen
Simulated site (any tab)	Sites → Site (+ time window)	Same objects as a live Site	Stable site_id; history assembled from contributing runs; provenance per evidence item; SIMULATED is provenance, not status

Replay	Site → Replay mode	Recorded Site history as-of t	Evidence by t, configuration valid at t, managed state as of t, no future leakage, no silent recomputation
Simulator Lab (Site View, Devices, Gateway, Scenarios, Runs)	Separate developer workspace	Simulated world / ScenarioDefinition / SimulationRun	Simulator owns truth and publication; shows no product conclusions; gateway output is the only simulator-produced information crossing into normal ingestion; runs are execution records (Draft / Committed), never Sites

10. Implementation acceptance tests

- Can a user understand what needs action without opening a chart?
- Can every visible claim be traced to evidence, configuration-at-time and decision logic?
- Does the screen distinguish measured, modelled, extrapolated, derived, scenario-based, observed-absence and unknown?
- Does every recommendation state how success will be verified?
- Can the simulator produce both the incident and a realistic non-incident look-alike?
- Does the financial view expose the technical bridge and avoid double counting?
- Does evidence coverage state the current claim ceiling and the next evidence that unlocks a stronger claim?
- Can a work order be closed without automatically closing the finding?
- Does a site remain useful when some sensors or records are absent by degrading confidence rather than synthesizing data?
- **New:** Can every site Findings / Work row be found, unchanged, in the corresponding global index with the same state? (No scoped-only states or counts.)
- **New:** Does clicking any object from a site tab change the left-nav highlight to that object's class while keeping the site in the breadcrumb?
- **New:** Is every coloured card an assessment, and does every numeric card state its basis?
- **New (v6.1):** Do Action, Incident and Maintenance detail pages all open with the same managed-object header, and does a task's state never contradict its provisional outcome?
- **New (v6.1):** Is source/gateway health rendered with the source vocabulary (Online / Stale / Offline) and never with the asset assessment vocabulary?
- Do a simulated Site and every object rooted in it carry site_id · SIMULATED · scenario · run provenance in the header and in exports, while showing no other difference in tabs, vocabulary or claim ceilings?
- Does the same world event authored with full / partial / missing evidence yield High / Medium confidence / Indeterminate, with no truth leakage into the product pipeline?
- Can Scenario Lab and the simulator be removed without changing any operator route, tab or object schema other than the Site mode/provenance fields?

Simulation integration acceptance (v6.4, replaces the v6.3 run-per-Site checks):

- **A. Declarative identity.** Given a Site YAML declaring site_id = maji-mini-grid-001, all gateway/source envelopes and downstream product artifacts (Evidence, Findings, Incidents, read models, exports, provenance) use that site_id.

- **B. Multiple runs, one Site history.** Given Run A → maji-mini-grid-001 → 00:00–12:00 and Run B → maji-mini-grid-001 → 12:00–24:00, requesting the Site for 00:00–24:00 exposes eligible evidence/artifacts from both intervals with preserved per-run provenance.
- **C. Run identity is not Site identity.** Changing run_id does not change the canonical Site identity or any Sites-index row.
- **D. Normal ingestion.** The Site page and analytics can be reproduced from serialized source envelopes without simulator runtime objects or SimulationTruth.
- **E. Truth isolation.** Changing private truth without changing observable evidence cannot alter normal AssetOps product conclusions.
- **F. Interactive causality.** A simulator intervention changes the simulated world/evidence path; any product consequence is derived downstream, never directly assigned.
- **G. Time-window consistency.** Changing the selected Site time window changes all time-dependent tabs/read models consistently according to their declared time semantics.
- **H. Overlap protection.** Ambiguous overlapping runs for the same Site/time interval cannot silently produce one canonical history.
- **I. Execution-speed invariance.** Given the same deterministic run identity defined in §3.6, executing at real time, 100× and maximum speed produces identical published envelopes and identical AssetOps conclusions; Fast-forward executes every intervening tick.
- **J. Boundary in both directions.** No operational Simulator Lab surface presents AssetOps Findings, assessments, confidence, reconciliation results or health scores as simulator outputs; private Expectations may contain explicitly labelled regression assertions about expected downstream behaviour, and those assertions never cross the ingestion boundary or influence product analytics. No gateway envelope, AssetOps evidence object, operator screen or export contains SimulationTruth (inside AssetOps it is visible only in the admin overlay).
- **K. Feature gate.** With simulator_lab.enabled = false, no simulator route, navigation item, entry point or truth overlay is reachable, while every operator route, simulated Site, its provenance and Replay behave exactly as before.
- **L. Interactive determinism.** A run reproduced from Site definition + scenario version + seed + simulator version + initial config/mappings + the ordered intervention log yields identical envelopes; the same identity minus the intervention log does not.
- Is AI limited to grounded summarization/navigation/drafting rather than sole factual finding generation?

11. Source basis and design authority

This revision is grounded in the AssetOps Product Experience, Site Lifecycle and Data Traceability Blueprint v3 for product architecture, site lifecycle, simulator behaviour, incident/work/verification, financial discipline and evidence readiness; in the AssetOps Source-to-Gateway Data Mapping and Product Traceability Specification for exact mini-grid/cold-room signals, derived evidence objects, screen consumption, quality, provenance and claim ceilings; in the v6 simulation review for the source-mode architecture, Scenario Lab, Replay and truth-vs-observed model; in the Simulator Lab mockup and reproduction description (v6.5) for the simulator workspace, time semantics and ingestion envelopes; in the frozen Simulation Integration Direction and ARCHITECT_GUIDE (v6.4) for Site identity, Site + time-window history, simulator setup, ingestion, protocol and persistence seams; and in v5/v6 of this document for the screen set, figures and example values, which are retained

wherever they were consistent. v6.1 applies the v6 review findings without changing the architecture.

The screen mockups intentionally do not add unsupported measurements or conclusions merely to make a layout look complete. Where the source contracts mark evidence as preferred, optional, missing or insufficient for a stronger claim, the UI retains that limitation.